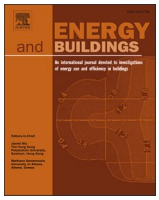
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An efficient hybrid deep neural network model for multi-horizon forecasting of power loads in academic buildings

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ABSTRACT

Accurate power consumption forecasting is crucial for optimizing energy use in smart buildings, improving efficiency and decision-making to enhance overall energy management. However, forecasting models encounter challenges in capturing complex load patterns and often focus on single-horizon predictions. Moreover, they lack advanced Deep Learning (DL) integration and overlook computational complexity, limiting adaptability across diverse environments. Motivated by existing research gaps, this study develops a novel hybrid DL model that integrates three blocks deploying Convolutional Neural Network (CNN), Long-Short-Term Memory (LSTM), and Multi-Layer Perceptron (MLP) with concatenation and residual connectivity to enhance forecasting accuracy and robustness. Our model consistently outperforms statistical, hybrid, baseline forecast, and deep learning models, achieving a Mean Absolute Error (MAE) of 5.93 and 5.93, Mean Squared Error (MSE) of 71.52 and 78.52, Root Mean Squared Error (RMSE) of 8.46 and 8.86, and Mean Absolute Percentage Error (MAPE) of 7.81% and 2.60% for the two distinct academic buildings, respectively. Moreover, the power consumption forecasting results in both buildings demonstrate that the proposed model reduces the computing time and memory usage up to 89.7% and 79.95%, respectively, for single-horizon forecasting and 91.9% and 96% for multi-horizon forecasting compared to traditional LSTM, BiLSTM, and GRU models. This paper demonstrates the benefits provided by advanced hybrid DL models in forecasting power consumption by capturing complex load patterns and temporal dependencies. Our findings emphasize the potential of single and multi-horizon forecasting to enhance energy efficiency, offering more reliable and adaptable solutions for smart building management.

1. Introduction

1.1. Motivation and forecasting methods

Digital technology is increasingly being deployed to organize and manage existing infrastructure, aiming to create smart cities where Information, Communication, and Technology (ICT) are integrated [1]. Sensors and actuators embedded into smart devices collect data from their surroundings, which can be leveraged for efficient decision-making on energy management for the built environment. Data from the World Watch Institute show that buildings are prime consumers of energy, accounting for 40% of global yearly energy consumption and 36% of overall carbon emissions, particularly in cities [2]. Higher education institutions and commercial buildings use more energy than residential buildings by 45% and 30%, respectively [3]. Implementing effective energy management systems and employing energy efficiency control tactics can reduce operational expenses by minimizing energy use and avoiding penalties, such as peak demand charges, power factor penalties, exceeding contracted demand, and overdue payment fees levied by utility companies [4]. Many countries have strengthened building codes to promote the use of energy-efficient building designs. These designs aim to decrease energy consumption, lower energy costs, and minimize environmental impacts such as pollution, carbon, and greenhouse gas emissions.

According to Amber et al. [3], the increasing demand for energy in buildings necessitates reliable load forecasting, which will promote effective planning, long-term strategies, and efficient measures to decrease carbon emissions and manage energy usage. Accurate consumption forecasting is important because it has the potential to reduce building energy usage by 10% to 30%. Power consumption forecasting is a crucial part of the energy management system, which provides essential information for the daily administration and planning of the electric utility grid. It enables building operators to make optimal energy management decisions and ensures the power system's safe and

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dependable operation. Research has demonstrated that improving the energy efficiency of buildings by developing precise and robust load forecasting models is an effective approach for managing energy, implementing demand response procedures, detecting faults, and establishing energy benchmarks [5].

Over the last two decades, a range of forecasting methods have been employed to anticipate the energy usage of buildings. These methods can be classified into three distinct categories: (i) engineering methods, (ii) statistical methods, and (iii) data-driven methods. Traditional engineering methods for forecasting power loads rely on physical and thermodynamic laws, necessitating complex building and environmental parameters. These techniques, such as EnergyPlus [6] and Ecotect, entail intricate simulations of building performance under different conditions, which are complex and time-consuming to implement. Statistical methods [7,8] frequently lack the precision and adaptability required for complex and dynamic systems, such as educational facilities. Statistical approaches are also inappropriate to solve non-linear features or adapt to significant changes in the data over time. As a result, these models face challenges when encompassing various exogenous variables and dealing with seasonality and irregularity. Although statistical approaches are effective in predicting short-term outcomes, they may not be reliable for long-term forecasts, requiring periodic recalibration and parameter adjustments.

To circumvent the limitations of engineering and statistical approaches, researchers integrate data-driven approaches [9,10,11]. Several research studies have used Machine Learning (ML) techniques [12] such as Decision Trees (DT), Support Vector Machines (SVM), K-Nearest Neighbor (KNN), and ensemble methods to learn the historical pattern of energy data and predict future energy consumption [13]. The ML technique can manage non-linear connections and integrating several inputs, such as weather and occupancy data, to enhance accuracy compared to conventional statistical methods [14]. However, the ML technique often necessitates substantial feature engineering, demands significant computer resources, and is unable to capture complex and long-term temporal dependencies effectively. More recently, neural networks have emerged as a tool in fields like image processing and data analysis [15]. To accurately forecast energy load, researchers now utilize deep learning techniques [16,17]. Most forecasting methods are based on Recurrent Neural Networks (RNNs) [18,11,19], transformer models [20], and Convolutional Neural Networks (CNNs) [21]. The Gated Recurrent Unit (GRU) network is a simpler and more effective variant of Long Short-Term Memory (LSTM) capable of addressing long-term dependency issues in RNNs and is currently a widely used network architecture. However, when managing long-term sequences, RNNs experience severe gradient disappearance, even with partial alleviation by LSTM. Additionally, RNN-based models necessitate serial calculation, resulting in unsatisfactory efficiency. CNNs [22] have achieved remarkable performance in computer vision, with well-known models such as AlexNet, ResNet [23], and DenseNet [24]. ResNet models have recently performed very well in sequence processing tasks, including speech synthesis, language modeling, machine translation, and energy demand forecasting. Deep CNNs (DCNN) with residual connectivity perform sophisticated operations that can learn discriminative features and nonlinear relationships, which benefit from extracting uncertainty factors from sequential data. Gated CNN incorporates the gated mechanism of LSTM into CNN. The gated mechanism is used to identify and evaluate information, resulting in superior results. DCNN excels in accurately calculating heating/cooling loads, total power consumption, and the operation and optimization of sub-level components.

1.2. Literature review

Engineering methods for building energy forecasting models include characterizing the dynamic thermal behavior of buildings on physical principles [25]. Alternatively, the building energy prediction models can be developed using statistical analysis, ML, and DL methods [26,27,28]. The statistical and data-driven methods utilize historical data to explore the relationships between variables [4,29,25]. Data-driven methods offer flexibility and efficiency in developing building energy forecasting models and have gained widespread attention. This section presents a review of recent advancements in data-driven building energy forecasting. Table 1 summarizes the relevant works based on the building type, energy load type, time interval and prediction horizon, proposed models, comparison models, key findings, as well as the limitations mentioned in the articles.

Fan et al. [30] compare the performance of data-driven models, including Multiple Linear Regression (MLR), Elastic Net (ELN), Random Forest (RF), Gradient Boosting Machines (GBM), Support Vector Regression (SVR), Extreme Gradient Boosting trees (XGB), and feed-forward Deep Neural Network (DNN), with different feature extraction methods in day-ahead building cooling load prediction. Their study on an academic building shows that deep auto-encoders can enhance the prediction performance, and the best performance is achieved by XGB using features extracted by them. Chou and Tran [31] propose multiple-phase hybrid prediction models to predict the electricity demand of a residential building. A comparative study shows that hybrid models outperform single and ensemble models. Seyedzadeh et al. [32] investigate several ML models, including Artificial Neural Network (ANN), SVM, Gaussian process (GP), RF, and Gradient Boosted Regression Trees (GBRT), for the prediction of buildings heating and cooling loads. The results show that the standard and advanced GBRT have the highest prediction accuracy. Almalaq and Zhang [33] present a short-term energy consumption prediction method for residential and commercial buildings based on evolutionary deep learning. The method combines a LSTM neural network with the Genetic Algorithm (GA) to optimize the time lags and the number of hidden neurons. Kim et al. [34] develop a Recurrent Inception Convolution Neural Network (RICNN) model for the multi short-term forecasting of building electric load. The model is examined for day-ahead electric load forecasting by using power usage data of three industrial distribution complexes, and the results show that the model yields higher accuracy than benchmark models.

Kim et al. [35] investigate the use of Autoregressive Integrated Moving Average (ARIMA) models, ARIMA-Generalized AutoRegressive Conditional Heteroskedasticity (ARIMA-GARCH) models, multiple seasonal exponential smoothing models, and ANN models for electricity load forecasting of institutional buildings. Their study identifies weather and holidays as key variables for prediction and shows that the ANN model with external variables performs best in short-term forecasting. Somu et al. [36] propose a DL-based framework for building energy consumption forecasting. Comparisons with several existing models, including ARIMA, Deep Belief Network (DBN), MLP, LSTM, and CNN, demonstrate the superior accuracy of their model. Syed et al. [37] develop a hybrid deep learning model to forecast the energy consumption of smart buildings. The model is shown to yield enhanced accuracy compared with other LSTM-based hybrid models, including CNN-LSTM, Convolutional LSTM (ConvLSTM), LSTM encoder-decoder model, stacked LSTM models, and an Averaging Regression Ensemble Model (AREM). Their results show that the LSTM-based RNN Multi-Input Multi-Output (MIMO) model achieves the best performance.

Moradzadeh et al. [38] propose a building heating load forecasting method based on cyber-secure federated deep learning. The method employs a global server to predict the heating load for local clients while preserving their privacy and without knowing their locations. Khan et al. [39] present an artificial intelligence-based framework for forecasting building power consumption and generation. Experimental results show that the model significantly reduces prediction errors compared to various state-of-the-art techniques. Morteza et al. [40] explore different RNN architectures for the prediction of building electricity and heating loads. Peng et al. [41] develop a CNN-LSTM network to predict the energy consumption of the Heating, Ventilation, and Air Conditioning (HVAC) system of buildings. Based on this prediction model, a deep reinforcement learning model is constructed to optimize the HVAC

Table 1
A summary of related works.

Articles	Building types	Load types	Time interval/ prediction horizon	Proposed models	Comparison models	Key findings	Limitations
Fan et al. [31]	educational	cooling	30 min/24 h	—	MLR, ELN, RF, GBM, SVR, XGB, DNN	deepauto-encoders enhance the prediction performance;XGB performs the best	—
Chou and Tran (Seyedzadeh et al., 2019)	residential	electricity	15 min/24 h	hybrid models	single models, ensemble models	achieve higher accuracy	—
Seyedzadeh et al. [33]	commercial, residential	heating, cooling	—	—	ANN, SVM, GP, RF, GBRT	GBRT is the most accurate; SVM suits low-dimensional data; ANN suits complex data	—
Almalaq and Zhang [34]	residential, commercial	electricity, energy	1 min/1 step, 5 min/1 step	GA-LSTM	ARIMA, DT, kNN, MLP, LSTM, GA-ANN	GA-LSTM performs the best	—
Kim et al. [35]	industrial	electricity	30 min/24 h	RICNN	MLP, CNN, RNN	RICNN yields the best accuracy and robustness	needs more training and inference time
Kim et al. [36]	institutional	electricity	1 h/24 h	—	ARIMA-basedmodels, ANN, exponential smoothing	weather and holiday are key variables for prediction; ANN with external variables performs the best	
Somu et al. [37]	academic	electricity	15 min/60 min	*k*CNN-LSTM	ARIMA, DBN, MLP, LSTM, CNN	improves accuracy	—
Syed et al. [38]	residential	electricity	10 min/1 step, 1 min/1 step	hybrid LSTM	CNN-LSTM, ConvLSTM, LSTM encoder-decoder, stacked LSTM, AREM	performs the best	—
Chalapathy et al. (Khan et al., 2023)	commercial	cooling	1 h/24 h	—	RNN models, SVR, XGBoost	RNN-MIMO performs the best	—
Luoand Oyedele [40]	educational	electricity	1 h/24 h	GA-LSTM	feedforward NN, LSTM	GA-LSTM performs the best	—
Jang et al. [41]	non–residential	heating	1 h/24 h	—	LSTM networks	LSTM with operation pattern data performs the best	limited building types for application; data quality and quantity
Moradzadeh et al. [42]	—	heating	1 day/—	cyber-secure federated DL	SVR, GRNN, LSTM, Bi-LSTM	shows advantages in forecastingaccuracy, data confidentiality, and generalizability	—
Khan et al. [43]	—	electricity	minutely, hourly, daily, weekly	ConvLSTM-BDGRU	various state-of-the-art techniques	significantly improves accuracy	—
Morteza et al.[44]	residential	electricity, heating	1 h/1h	RNN	SVM, GB	enhances accuracy	—
Peng et al. [45]	—	HVAC	—	CNN-LSTM	GRU, LSTM, CNN	performs the best	—
Lin et al. [46]	non–residential	electricity	1 h/24 h	predictive power demand analytics	3 LSTM architectures	single-encoder decoder model outperforms standard and multi-encoder decoder models	—
Wang et al. [47]	community	electricity	1 h/96 h	adaptive federated learning	RNN, LSTM, GRU	improves anomaly prediction performance	sensors noise not considered; scalability not investigated sufficiently;

(*continued on next page*)

Table 1 (*continued*)

Articles	Building types	Load types	Time interval/ prediction horizon	Proposed models	Comparison models	Key findings	Limitations
Wang et al. [48]	commercial	electricity, temperature	1 h/24 h	multi-task DL	LSTM, CNN, transformer	achieves superior performance across all tasks	benchmark not established high computing resources consumption; large training time
72							

energy consumption.

In some works, the energy prediction model integrates other modules alongside the load forecasting module, enhancing its capability to perform multiple tasks, such as anomaly detection. For example, Lin et al. [42] propose a predictive power demand analytic methodology based on deep learning and symbolic aggregate approximation. The experimental results show that the model achieves good accuracy in forecasting and anomaly prediction, and the single-encoder decoder LSTM network outperforms standard and multi-encoder decoder models. Wang et al. [43] develop an adaptive federated learning framework for forecasting the energy load of community buildings. The framework addresses data isolation due to privacy regulations and integrates an anomaly prediction module based on Gaussian mixture models.

1.3. Research gaps

The State-Of-The-Art (SOTA) in power consumption forecasting often relies on statistical or traditional machine learning models. For example, [13,32,8,35] adopted different ML algorithms and statistical approaches, such as ARIMA, ARIMA-GARCH, Logistic Regression (LR), SVM, Naive Bayes (NB), DT, KNN, and Neural Networks (NNs) for load forecasting. These methods often struggle to capture the complex, nonlinear relationships inherent in energy consumption data, especially in dynamic environments like academic buildings where occupancy, class schedules, weather, and HVAC systems influence load patterns. Many of these techniques also fail to handle multivariate data effectively, have limited ability for multi-horizon forecasting, and may exhibit poor generalization when applied to diverse building types or under varying operational conditions. Additionally, they tend to require extensive data preprocessing and are sensitive to the choice of hyperparameters, making them less adaptable for real-time, large-scale energy forecasting applications. Most deep learning-based SOTA models primarily focus on single-horizon forecasting. For example, studies such as [44] and [45] have utilized deep learning techniques, including TCN and RNN, to perform one-step-ahead predictions. This is primarily due to the increased complexity and computational demands associated with predicting multiple future time points. Inteha et al. [46] perform day-ahead forecasting implementing LSTM, GRU, and bidirectional gated recurrent unit. This model's limitation is that it is designed for 24-hour forecasts. While its performance is extended to 48-hour intervals for evaluation, its generalizability and effectiveness beyond this time frame remain unevaluated. Nonetheless, all these aforementioned SOTA models simply stacked the sequential layers that are less efficient and computationally expensive. Multi-horizon forecasting requires the model to capture and accurately predict a range of dependencies and interactions over various future intervals, significantly increasing the task's complexity. Additionally, ensuring accuracy across multiple horizons simultaneously is challenging and often leads to higher computational costs and inefficiencies, making it less explored than single-horizon forecasting. From the discussion above, five key research gaps in the SOTA models are listed as follows: (i) dealing with complex load patterns in academic buildings, such as the occupancy (i.e., number of occupants, schedules, status, and diverse operations), and seasonal variations create highly variable and unpredictable energy consumption patterns; (ii) limited focus on multi-horizon forecasting; (iii) integration of advanced deep learning techniques, such as feature re-usability and transformation via residual connection, concatenation, and hybrid network in the forecasting model; (iv) lack of computational complexity analysis, and (v) adaptability across diverse academic buildings.

2. Novelty and contributions

This study proposes an efficient hybrid deep neural network for power load forecasting to address the shortcomings of the existing SOTA models. The novelty of our proposed model lies in its comprehensive hybrid architecture. Existing forecasting models have yet to employ advanced deep learning techniques, such as deep CNN with residual connectivity and concatenation and the combined integration of CNNs, LSTM, and Multilayer Perceptron (MLP). The proposed model combines the strengths of CNN for feature extraction, MLP for nonlinear transformation, and the LSTM layer for capturing temporal dependencies. This fusion enables the model to efficiently capture both linear and nonlinear dependencies and overcome the nonlinearity issue of power load patterns. Furthermore, we introduced residual connections and feature concatenation, enhancing feature reusability and transformation within the network, which allows the model to handle dynamic and unpredictable consumption patterns. This high-level architecture offers high prediction accuracy while maintaining low computational expense. The model's architecture is also designed to handle single-step and multi-step ahead predictions, addressing the limitation of existing models that often focus on single-horizon predictions. This dual capability allows it to predict immediate energy consumption and future consumption patterns. This extension is particularly valuable for energy management applications, where accurate predictions over longer horizons are essential for optimizing resource allocation. In addition, a key innovation of our work is computational complexity analysis, which evaluates learnable parameters, memory size, and inference time. This approach provides a clearer understanding of the trade-offs between prediction accuracy and computational cost, which has not been sufficiently addressed in existing research. Finally, we demonstrate the generalizability of our approach by applying it to two different academic building datasets, showcasing its robustness and adaptability to various real-world scenarios.

The following list summarizes our contributions in this work:

- We developed a hybrid deep learning model deploying CNN, LSTM, and MLP layers. The proposed model is composed of three key blocks, named Conv-block, LSTM-block, and MLP-block. These blocks are designed with convolution, LSTM, and MLP layers incorporating residual connectivity and a concatenation layer that can efficiently handle the complex load patterns inherent in academic buildings. This design configuration of the proposed model enhances forecasting accuracy and robustness with low computational expense.

- We conducted comprehensive evaluations focused on single-horizon and multi-horizon forecasting, providing insights into the model's performance over extended periods.
- We validated the proposed model on two different building datasets, demonstrating its adaptability and effectiveness across varied educational building environments.
- We evaluated the robustness of the proposed model through simulations incorporating, among other considerations, the Mean Absolute Error (MAE), Mean Squared Error (MSE), Root Mean Squared Error (RMSE) and Mean Absolute Percentage Error (MAPE). Moreover, the computational complexity was evaluated in terms of learnable parameters, memory size, and inference time. The proposed network is evaluated by comparing it with statistical, baseline, and four different deep learning models. The proposed model consistently outperformed, highlighting its accuracy, complexity, and robustness in power consumption forecasting.

The rest of this article is organized as follows: Section 2 briefly describes the research methodology, including the system model, dataset description and preprocessing, problem formulation and a comprehensive explanation of the proposed hybrid deep learning model. Section 3 presents the experimental results and discussion. Finally, Section 4 concludes the paper.

3. Methodology

This section explains the system model for power forecasting, introduces power consumption datasets used in this work, and discusses data analysis and preprocessing methodologies. Moreover, it also includes the methodology for problem formulation and the development of the proposed deep learning model.

3.1. System architecture

Fig. 1 illustrates the proposed system model. The proposed system architecture for power forecasting in academic buildings aims to efficiently collect, process, and analyze data to develop robust forecasting models.

Each layer in the architecture performs a specific function to ensure accurate and reliable predictions. The architecture starts with the physical layer, which involves gathering energy data from 152 campus buildings using the SkySpark platform. This platform collects data from various sensors and electric meters installed across the buildings, capturing real-time power consumption information. The Sensing Layer facilitates this data collection by using a variety of sensors and electric meters to record real-time energy consumption data. This layer provides granular data, which is crucial for detailed analysis and accurate forecasting. The cloud server processes and stores the collected data. This layer manages the intake of substantial amounts of data, guaranteeing its efficient storage and simple accessibility for subsequent processing and analysis. Skyspark has a data visualization engine that displays the real-time power consumption of buildings, which is publicly accessible in CSV format. The collected data then undergoes several preparation procedures in the preprocessing layer, including data cleaning, missing value imputation, outlier detection, and normalization. After preprocessing, the data splitting layer divides the data into training and testing sets, crucial for evaluating the forecasting models' performance. The model building layer explores statistical and DL techniques, such as ARIMA, Prophet, LSTM, GRU, BiLSTM, CNN, and the proposed hybrid DL model. Finally, the performance evaluation layer checks the models using performance metrics including MSE, RMSE, MAPE, and MAE. It also checks the models' computational complexity by calculating memory usage, learnable parameters, and inference time. This structured approach ensures the creation of accurate and reliable power forecasting models tailored to the unique consumption patterns of academic buildings.

3.2. Dataset description and analysis

This study conducts experiments using a real-world power consumption dataset. We collected hourly power consumption data from the Centre for Interactive Research on Sustainability (CIRS) and the University Services Building (USB) on the University of British Columbia campus from January 1, 2019, to December 31, 2023. The data was collected from a SkySpark database. The University of British Columbia implemented the SkySpark platform as part of their Campus as Living Laboratory initiative to advance research in building performance and energy management systems. We present the data as discretized one-hour meter readings that show the total power consumption (kWh) recorded within each interval. The data timestamps range from January 1, 2019, to December 31, 2023, and each building has 43,842 hourly (1825 days) sampled data points.

3.3. Data preprocessing

In practical situations, power consumption data is prone to several inconsistencies, such as missing data, incomplete data, noise, and other factors, which might result in subpar data analysis. Below, we provide a comprehensive explanation of the methods employed to preprocess the power consumption data.

3.3.1. Missing values

Handling missing values is a crucial step in the data preprocessing phase for power consumption data. Missing data can arise due to several reasons including sensor malfunctions, data transmission errors, or maintenance activities. These gaps in the data seta can significantly impact the performance of machine learning and deep learning models if not addressed properly. Therefore, this research employs strategies to manage and impute missing values effectively.

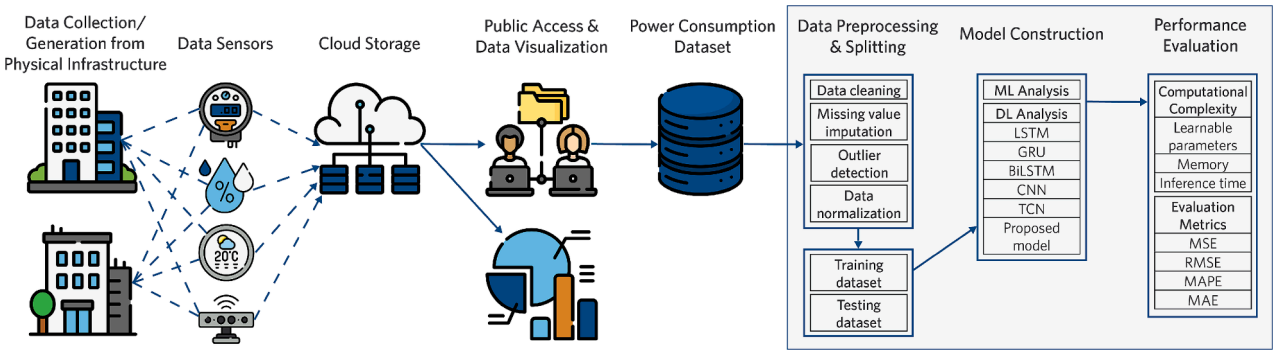


Fig. 1. Illustration of the proposed system model for building power load forecasting.

3.3.2. Outlier detection

Identifying outliers is an essential part of the data preprocessing stage, particularly for power consumption data. Anomalies in this type of data can have a substantial effect on the accuracy and effectiveness of forecasting models. To detect outliers, we employed the Interquartile Range (IQR) method, which is a robust and widely used technique for identifying outliers in a dataset. The IQR approach entails determining the interval in which the middle 50% of the data points are located. Next, we identify the outliers as data points that fall outside this range. Finally, we impute these outliers and replace them with the median value. We ensure a methodical approach to identifying and managing outliers by utilizing the IQR method, thereby improving the quality and dependability of the power consumption data.

3.3.3. Data normalization

Due to the wide range of values in the power consumption data, we standardize the data using the min–max normalization technique to prevent erroneous projections. Min-max normalization is a data preparation method that involves scaling a dataset's features to a specific range, usually between 0 and 1. This approach guarantees that all features have an equal impact on the analysis, hence preventing features with wider ranges from overpowering those with lower ranges. The equation for min–max normalization is as follows:

$$X' = \frac{X - X_{min}}{X_{max} - X_{min}}. \tag{1}$$

where X' corresponds to normalized numeric results; X is the original value, where X_{max} and X_{min} represent the maximum and minimum values of the feature, respectively. Min-max normalization helps enhance the convergence speed of gradient-based algorithms and guarantees that the model treats all features equally.

3.4. Problem formulation

In a building with multiple features, the problem of power consumption forecasting encompasses the total power consumed by various components (such as plugs, lights, fans, air conditioners, servers, computers, etc.) and environmental factors (such as humidity, temperature, etc.). These components and factors are monitored and recorded by sensors (such as smart meters, temperature sensors, etc.) that are installed at distinct levels of the buildings. The power consumed at timestamp t can be expressed as in Eq. (2):

$$P_t = \left\{P_t^1, P_t^2, ..., P_t^i, ..., P_t^n\right\}, \tag{2}$$

where P_t^i represents the power consumption recorded by the i-th sensor at the given time stamp t. This study employs a MIMO sliding window technique to forecast power consumption more accurately. The input window size S_i and output window size S_o can be presented as $\{S_i, S_o\} \in N$. Further, we can calculate the total number of input and output windows as follows:

$$T_{io} = (N_s - S_i - S_o + 1)/d, \tag{3}$$

here N_s and d represent the number of power consumption samples and the output intervals, respectively. The input window I_w of size S_i and the output window O_w of size S_o can be presented as follows:

$$I_w = \left\{P_t, P_{t+1}, ..., P_{t+S_i-1}\right\}, \tag{4}$$

$$O_w = \left\{P_{t+S_i}, P_{t+S_i+1}, ..., P_{t+S_i+S_o-1}\right\}, \tag{5}$$

In this study, the DL and our proposed model are designed as power forecasters that employ an approximation function to process the input window samples I_w and generate O_w as follows:

$$O_w = f(I_w), f : P^{S_i} \rightarrow P^{S_o}. \tag{6}$$

Eq. ([Eq_6]) demonstrates that given an input window (S_i), the model (f) learns to forecast the power consumption values of the output window (S_o). The minimum forecasting error F_{Error} for the power consumption can be presented as:

$$F_{Error} = \frac{1}{n}\sum_{i=1}^{n}\left|P_i^t - P_t'\right|. \tag{7}$$

where, P_i^t and P_t' are the actual and forecasted power consumption value of the ith sensor at t^{th} timestamp.

3.5. Proposed model architecture for power load forecasting

Forecasting power consumption in academic buildings presents several unique challenges due to the distinct characteristics of these facilities. Academic buildings often experience highly variable occupancy patterns influenced by class schedules, exams, holidays, and special events, leading to significant temporal fluctuations. Additionally, academic buildings contain diverse spaces such as classrooms, laboratories, libraries, and auditoriums, each with different power usage patterns. Extended operational hours, including evenings and weekends for research activities, further complicated power prediction. Alongside these specific challenges, broader uncertainties also impact power forecasting in academic buildings, such as fluctuating weather conditions and the general disparity in power usage between weekdays and weekends. The diverse activities, occupancy patterns and uncertainties within educational facilities add further complexity to accurate power load prediction. Due to the advanced learning capabilities, the DNN can automatically extract the key features from historical load sequences and forecast power load accurately. However, the traditional sequential deep learning models, while capable of achieving excellent accuracy, often incur high computational costs, making efficient and precise forecasting even more difficult. To solve the issues discussed, we propose a novel deep network structure that leverages a combination of CNN, LSTM, and MLP layers with advanced deep learning concepts. The key goal of this architecture is to attain an elevated level of precision while keeping computational costs to a minimum. This section provides a detailed explanation of the model's structure, including the specific layers utilized, their mathematical representations, and how they contribute to the overall performance of the model.

As illustrated in Fig. 2, the proposed model initializes with two regular convolution layers paired with a ReLU layer. The purpose of these layers is to extract generic features. In addition, the proposed model comprises three key blocks, namely Conv-block, LSTM-block, and MLP-block. First, the input power load sequence data was preprocessed using the MinMax normalization technique. The input layer receives the time series data in the shape $I \in N^{batchsize \times 96 \times 1}$. We incorporate two 1-dimensional convolution layers to extract the temporal features of the input time series data. Initially, a 1×3 convolution layer cascaded with another 3×1 convolution layer. Both convolution layers have an 8 number of kernels with a stride of $(1,1)$. Convolution layers are powerful tools for learning spatial hierarchies in data. They are particularly effective at extracting local patterns from time-series data, such as the fluctuating power consumption patterns in buildings. The key concept behind using CNNs in this context is that the convolutional layers can automatically identify important features in the input data, such as trends and seasonal variations, through local receptive fields. By applying multiple convolutional layers, the model can progressively extract more complex features. The feature extraction mechanism of the convolution layer is theoretically described as below:

To comprehend regular convolution more effectively, consider $\psi_{x,y}^n \in \mathbb{R}^{K_h \times K_w}$ as the collection of kernel weights. Here, n represents the n th feature map of the input feature volume, and the variables indexing the kernel are denoted as (x, y). The convolution operation between the filter and the n th feature map of the input $I \in \mathbb{R}^{H_i \times W_i}$ is achieved by summing the products of the kernel weight and input feature map. The output feature volume is obtained by adding up the n th products using

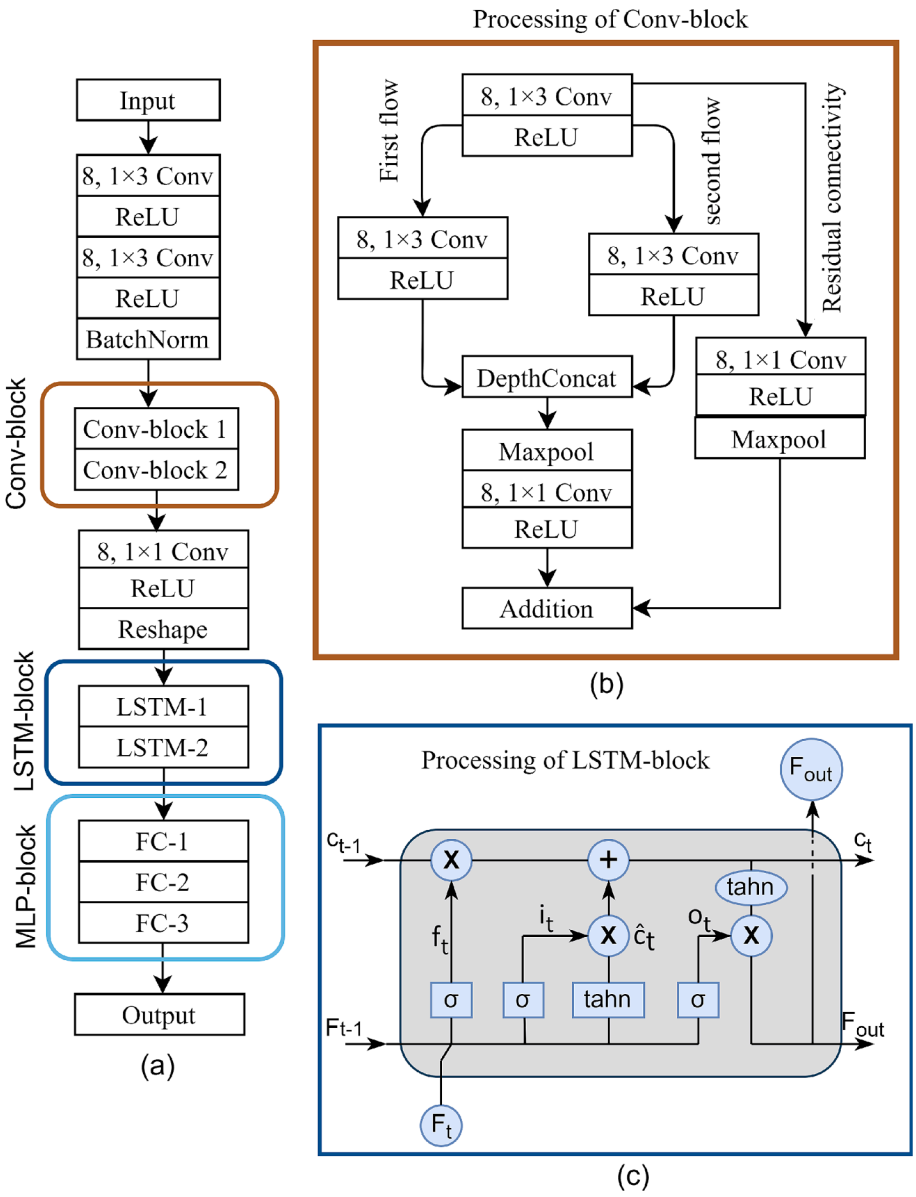


Fig. 2. Illustration of the proposed DL model for building power load forecasting: (a) proposed hybrid model including Conv-block, LSTM-block, and MLP-block, (b) architecture of Conv-block, and (c) structure of the LSTM-block.

the convolution approach. The convolution operation O is defined in the spatial position (p, q) as follows:

$$O_{p,q} = \sum_{x=1}^{K_h} \sum_{y=1}^{K_w} \psi_{x,y} I_{(x+p),(y+q)}, \tag{8}$$

We can express the convolution output, represented as $V_{p,q}$ at the spatial coordinates (p, q), as follows incorporating the bias term b as:

$$V_{p,q} = O_{p,q} + b, \tag{9}$$

Following the previously discussed general feature extraction strategy for the convolution layer, we can mathematically present the extracted feature map Y from the cascaded convolution layer as follows:

$$\mathrm{Y} = \mathscr{F}_{1\times3}(\mathscr{F}_{3\times1}(I)), \tag{10}$$

here I is the input set of power load that is fed into the 1×3 and 3×1 convolution layers, which are processed and generate the output features presented as $\mathscr{F}_{1\times3}$, and $\mathscr{F}_{3\times1}$, respectively. Importantly, the ReLU activation function follows each of the convolution layers in the proposed architecture for non-linearity. The ReLU layer is efficient due to its capability of mitigating overfitting problems. The ReLU function outputs zero for any input value less than zero and passes the input value unchanged if it is equal to or greater than zero. The operation of the ReLU can be expressed as follows:

$$\mathrm{ReLU}(Y) = \begin{cases} Y \; if \; Y \geqslant 0 \\ 0 \; if \; Y \geqslant 0 \end{cases} \tag{11}$$

The first Conv-block processes the output feature from the ReLU layer. The Conv-block consists of three 1×3, two 1×1 convolution layers, two maxpool layers, one concatenation, and one residual connectivity. Firstly, 1×3 convolution layer receives the output features from the ReLU layer and extracts features with eight kernels. The output feature maps $\mathrm{X}_{1\times3}$ of this layer can be presented as,

$$\mathrm{X}_{1\times3} = \mathscr{F}_{1\times3}(\mathrm{Y}), \tag{12}$$

The output features then pass through three flow feature extractors to capture the most discriminative temporal features. The first and second flows are employed with two 1×3 convolution layers, including

32 kernels with the stride of $s(1,1)$. The concatenation layer combines the aggregated features from these parallel convolution layers. Feature concatenation is a crucial technique used in our model, particularly within the Conv-block. This method combines outputs from different layers and enables parallel feature abstraction, allowing the model to capture multi-scale features. This capability is essential for effectively addressing the complex and dynamic load patterns observed in building power consumption. Concatenation works by joining multiple feature maps along the depth axis, which allows the network to merge information from various levels of abstraction. In our model, concatenating feature maps from different convolutional layers enables it to capture both local (short-term) and global (long-term) dependencies in time-series data simultaneously. This multi-scale approach ensures that the model learns varying levels of granularity, such as short-term fluctuations in power consumption and broader trends over extended periods. By employing concatenation, the model can retain potentially important features that might be overlooked in a simpler architecture. Retaining information from multiple layers enhances the model's ability to generalize and adapt to diverse consumption patterns across different buildings and time horizons. The output feature maps of the concatenation layer $\mathsf{F}^{\mathrm{cb}_1}_{\mathrm{concat}}$ can be presented as follows:

$$\mathsf{F}^{\mathrm{cb}_1}_{\mathrm{concat}} = \mathscr{D}(\mathscr{F}_{1\times3}(\mathsf{X}_{1\times3}), \mathscr{F}'_{1\times3}(\mathsf{X}_{1\times3})), \tag{13}$$

where $\mathscr{D}$ represents the concatenation function, and $\mathscr{F}_{1\times3}(\mathsf{X}_{1\times3})$, and $\mathscr{F}'_{1\times3}(\mathsf{X}_{1\times3})$ represent the first and second flow feature maps from both 1×3 convolution layer, respectively.

For the dimension reduction of the concatenated output, we imposed the maxpool layer with pool size 2. The maxpool layers offer reduced computational expense with better learning efficiencies that increase power load forecasting accuracy with low computational expense. A unit convolution layer then expands the maxpool layer's decreased output feature volume at a low computational cost. We can present the feature abstraction of the first flow in the initial convolution block $\mathsf{F}^{\mathrm{cb1}}_{\mathrm{Flow}_1}$ as follows:

$$\mathsf{F}^{\mathrm{cb1}}_{\mathrm{Flow}_1} = \mathscr{F}_{1\times1}\left(\mathscr{F}_{\max}\left(\mathsf{F}^{\mathrm{cb}_1}_{\mathrm{concat}}\right)\right), \tag{14}$$

where, $\mathscr{F}_{\max}$ and $\mathscr{F}_{1\times1}$ present the output features of the maxpool and point convolution layers, respectively.

In parallel, a point convolution layer and a maxpool layer work as a skip connection that aggregates the features from the previous 1×3 convolution layer's outcome. The feature map from the skip connection $\mathsf{F}^{\mathrm{cb1}}_{\mathrm{Flow}_2}$ can be shown as follows:

$$\mathsf{F}^{\mathrm{cb1}}_{\mathrm{Flow}_2} = \mathscr{F}_{1\times1}(\mathscr{F}_{\max}(\mathsf{X}_{1\times3})), \tag{15}$$

Skip connection, also known as residual connectivity, adds the extracted features from the $\mathsf{X}_{1\times3}$ and the concatenated features from the first and second flows in the Conv-block. By incorporating the residual connection, the model can effectively capture a diverse array of temporal properties and interactions, enhancing the richness of the learned representations. It enhances the learning capabilities of deeper networks by facilitating the flow of gradients during the backpropagation process. In traditional deep networks, as the depth increases, the gradients can become very small, leading to the vanishing gradient problem. This makes it challenging for the network to learn complex features. This challenge is particularly significant in time-series forecasting tasks, where capturing long-range temporal dependencies is essential. The core idea behind residual connections is the use of skip connections, where the output of a layer is directly added to the input of a deeper layer. This structure enables the model to learn identity mappings, making optimization easier and preventing the loss of important features. In our model, residual connections are utilized to combine feature maps from various convolutional layers, ensuring that both shallow and deep features are retained and effectively propagated throughout the network. This approach helps maintain the integrity of the learned representations, especially when dealing with high-dimensional time-series data that may be sparse or noisy. By implementing residual connections, our model reduces the risk of overfitting and enhances the stability of the learning process. Additionally, these connections accelerate convergence during training, allowing the model to achieve improved performance with fewer epochs. Mathematically, the feature of the residual connection $\mathsf{F}^{\mathrm{cb1}}_{\mathrm{add}}$ can be shown as follows:

$$\mathsf{F}^{\mathrm{cb1}}_{\mathrm{add}} = \mathsf{F}^{\mathrm{cb1}}_{\mathrm{Flow}_1} + \mathsf{F}^{\mathrm{cb1}}_{\mathrm{Flow}_2}. \tag{16}$$

where, $\mathsf{F}^{\mathrm{cb1}}_{\mathrm{Flow}_1}$ and $\mathsf{F}^{\mathrm{cb1}}_{\mathrm{Flow}_2}$ refer to the feature maps from the first and second flows of the first Conv-block, respectively. We assigned another Conv-block to extract the most disentangled features from the first block's output, adhering to the same design constraints as the first Conv-block.

To stabilize the forecasting accuracy, the proposed architecture includes an LSTM-block that consists of two LSTM layers incorporating 32 neurons. Power consumption is inherently dependent on time, with these dependencies varying across multiple time scales. Therefore, it is crucial for the model to understand how past consumption influences future patterns. LSTM units are designed to combat the vanishing gradient problem, allowing the model to effectively learn long-range dependencies. LSTMs are specifically tailored for learning from sequences. Unlike traditional feedforward networks, LSTMs maintain a memory of past information through gating mechanisms. These mechanisms regulate the flow of information, enabling the network to selectively forget or retain important details over extended periods. This characteristic makes LSTMs particularly well-suited for time-series forecasting, where grasping the temporal context of input data is essential for predicting future consumption patterns. In the subsequent paragraph, we explained how the LSTM block contributes to extracting temporal features of the power consumption dataset. Before implementing the LSTM-block, the output features of the second Conv-block are expanded and reshaped with the 1×1 convolution layer and the reshape function that can be shown as follows:

$$R_t = \mathscr{F}_{reshape}\left(\mathscr{F}_{1\times1}\left(\mathsf{F}^{\mathrm{cb2}}_{\mathrm{Out}}\right)\right), \tag{17}$$

where, $\mathsf{F}^{\mathrm{cb2}}_{\mathrm{Out}}$, $\mathscr{F}_{1\times1}$, $\mathscr{F}_{reshape}$, and R_t present the output features map of the second Conv-block, 1×1 convolution layer, the reshape function, and the rescaling features that will be input to the LSTM-block, respectively.

The LSTM-block plays a crucial role in capturing long-term dependencies and temporal patterns in the time series power consumption data in the proposed model. The first LSTM layer returns sequences to retain the temporal structure, while the subsequent layer produces the final forecasted values. An LSTM layer consists of many LSTM cells, each comprising three key gates: the forget gate, input gate, and output gate. These gates regulate the flow of information from the previous blocks and play an important role in maintaining the cell state, which serves as the network's memory. Firstly, the forget gate receives the reshaped features of the power load consumption R_t and the previous cell output R_{t-1}, and then processes them through a sigmoid gate σ to eliminate unnecessary features. Then, the acquired data is consolidated through the multiplication operation. The expression for the output of the forget gate F_t function is as follows:

$$F_t = \sigma\left(M_{wf}.[R_{t-1}, R_t] + b_f\right), \tag{18}$$

where, M_{wf} and b_f present the weight matrix and the offset of the forget gate, respectively.

Subsequently, the input gate selects the essential features that need to be included in the cell state. Two functions carry out this operation. For instance, the sigmoid function manages the input features and determines which values require updating to the current state. Furthermore, the input gate utilizes a hyperbolic tangent function (tanh) to produce a vector representing the information that will be incorporated

into the current state. The following expression describes the input gate I_t and cell state processing $\widetilde{C}_t$ using sigmoid and tanh function is as follows:

$$I_t = \sigma(M_{wi}.[R_{t-1}, R_t] + b_i), \tag{19}$$

$$\widetilde{C}_t = \tanh(M_{wc}.[R_{t-1}, R_t] + b_c), \tag{20}$$

where, M_{wi}, M_{wc}, b_i, and b_c present the weight matrix, the offset of the input gate, and the cell state, respectively. The previous cell state is merged with the new candidate value to update the cell state. This is done by multiplying the previous cell state C_{t-1} with the forget gate state F_t. Then, I_t is multiplied by $\widetilde{C}_t$ and added to form the new cell state C_t can be shown as follows:

$$C_t = F_t \times C_{t-1} + I_t \times \widetilde{C}_t, \tag{21}$$

Finally, the output gate is a key feature aggregator that determines the value of the following hidden state. This hidden state is then utilized for the subsequent time step and serves as the output of the LSTM cell for the current time step. The mapping function of the output gate O_t and the outcome of LSTM block F_{lstm} can be shown as follows:

$$O_t = \sigma(M_{wo}.[R_{t-1}, R_t] + b_o), \tag{22}$$

$$F_{lstm} = O_t \times tanh(C_t). \tag{23}$$

In our proposed model, the first LSTM layer passes its hidden state to the second LSTM layer, which further refines the sequence representation for power load forecasting. The output of the second LSTM layer is then used for the final prediction. This allows the model to capture intricate temporal relationships in the power load data, leading to higher forecast accuracy.

Following the LSTM layers, the output is channeled into the crucial MLP-block. MLP-block is employed to introduce non-linearity into the model. After the CNN layers extract important features and the LSTM layers capture temporal dependencies, the MLP layers apply nonlinear transformations to these features. This allows the model to learn complex relationships between input variables that linear models may fail to capture. The use of MLPs in our hybrid model enhances the model's ability to map the learned features to the final forecasting output. This block plays a pivotal role in refining the features and generating the final power load predictions. The MLP-block, a key component of our model, is composed of three dense layers with 32, 16, and 48 neurons, respectively. These layers are further enhanced by the ReLU activation function, which significantly contributes to the model's performance. The working process of the MLP-block can be shown as follows:

$$\mathrm{F}_{\mathrm{MLP}}^{\mathrm{d1}} = ReLU(W_{d1}.F_{lstm} + b_{d1}), \tag{24}$$

$$\mathrm{F}_{\mathrm{MLP}}^{\mathrm{d2}} = ReLU\left(W_{d2}.\mathrm{F}_{\mathrm{MLP}}^{\mathrm{d1}} + b_{d2}\right), \tag{25}$$

$$\mathrm{F}_{\mathrm{MLP}}^{\mathrm{out}} = ReLU\left(W_{d3}.\mathrm{F}_{\mathrm{MLP}}^{\mathrm{d2}} + b_{d3}\right). \tag{26}$$

here, W_{d1}, W_{d2}, W_{d3}, b_{d1}, b_{d2}, b_{d3}, $\mathrm{F}_{\mathrm{MLP}}^{\mathrm{d1}}$, $\mathrm{F}_{\mathrm{MLP}}^{\mathrm{d2}}$, and $\mathrm{F}_{\mathrm{MLP}}^{\mathrm{out}}$ present the weight matrix, bias, and the output features of the first, second, and the final dense layer in MLP-block, respectively. Notably, the final output layer, equipped with 48 neurons, produces the final prediction of future power loads. Finally, this layered approach, combining the depth and temporal feature extraction capability of CNN and LSTM with the nonlinear processing power of MLP, ensures high accuracy in power load forecasting while maintaining computational efficiency.

3.6. Computational complexity of the proposed approach

The complexity of the proposed hybrid DL model arises from its architecture, which integrates various components for effective feature extraction, sequence modeling, and prediction. This study investigates three key computation-assessing factors of DL models, such as the number of learnable parameters, memory size, and inference time, to assess the computation complexity of the proposed network. The total number of learnable parameters in the model is determined by the convolutional layers, LSTM layers, and dense layers. The proposed model consists of the 1-dimensional convolution layer. Each convolution layer has parameters proportional to the product of the filter size, the number of input channels, and the number of output filters. The computation of the learnable parameter of the convolution layer can be shown as follows:

$$P_{conv} = (F_s \times I_i + 1) \times I_o, \tag{27}$$

here, P_{conv} is the learnable parameter for each convolution layer, where F_s, I_i, and I_o is the filter size, input and output channel, respectively. The LSTM layers significantly contribute to the learnable parameters, including weights for input, forget, output gates, and cell state. For an LSTM layer with n units and m input features, the total number of parameters, P_L is:

$$P_L = 4 \times [n \times (m + n) + n], \tag{28}$$

Each dense layer contributes parameters proportional to the product of the input size I_n and output size I_{out}, plus biases b. The computation of the learnable parameter of the dense layer can be shown as follows:

$$P_d = (I_n \times I_{out}) + b. \tag{29}$$

Memory size is another computational complexity measurement factor that is influenced by several factors, including the size of the input data, the number of model parameters, and the intermediate activations that occur during both the forward and backward passes. The proposed model utilizes max-pooling layers to reduce the dimensionality of feature maps, which minimizes memory overhead while still preserving essential features. Furthermore, incorporating dropout regularization during training helps to reduce the risk of overfitting without significantly increasing memory requirements. However, the inference time of the model depends on the architecture's depth and the operations performed at each layer. Convolutional layers provide efficient parallelization on GPUs, thereby significantly accelerating feature extraction. In contrast, LSTM layers are fundamentally sequential, which may lead to bottlenecks during inference. To address this issue, the architecture is optimized by reducing the LSTM sequence length after pooling and incorporating computationally efficient dense layers. Following the aforementioned consideration, the proposed model has 20,257 and 21,056 learnable parameters; a memory size of 79.13 KB and 84.17 KB; and an average inference time of 1.8 and 1.4 s for single-horizon and 48-horizon forecasting, respectively, for both the CIRS and USB building datasets.

4. Simulation results and analysis

This section presents the performance of the proposed hybrid neural network framework through comprehensive evaluations. The high computational costs associated with advanced forecasting models, including deep learning techniques, usually result in significant demands on memory and execution time. Therefore, this work focuses on achieving high forecasting accuracy while reducing computational complexity by adopting an efficient hybrid neural network architecture. The performance evaluation of the proposed framework is conducted through four experiments, each varying in different measurements as follows:

- In the first experiment, we evaluate the single step forecasting performance using statistical approaches, including Seasonal ARIMA (SARIMA), Prophet, Neural Prophet, Persistence, and Mean Forecast, on the CIRS building dataset.

- The second experiment involves a comparative analysis of five different deep learning techniques for both single-step and multi-step forecasting using the CIRS building power consumption dataset.
- In the third experiment, the proposed model is validated using the USB building dataset for both single and multi-horizon forecasting, ensuring the robustness and generalizability of the model across different datasets.
- Lastly, the computational complexity is measured in terms of learnable parameters, memory size, and inference time for single and multi-horizon for both CIRS and USB building datasets.

4.1. System configuration

We used a robust computational setup to execute the simulations for power consumption forecasting. The software environment included Jupyter Notebook, which was managed through Anaconda, leveraging Python's capabilities for implementing deep learning algorithms. Neural network models were implemented using the Keras framework, utilizing a TensorFlow backend. The hardware platform featured an Intel (R) Core(TM) i7-10750H CPU running at 2.60 GHz, with a total of 16.0 GB RAM to ensure smooth execution of the simulations. Additionally, we used an RTX2080 GPU to speed up the deep learning computations, which improved the simulation tasks' efficiency and performance.

4.2. Evaluation metrics

To assess the effectiveness of the proposed and other forecasting models, we employ four widely used metrics namely MAE, MSE, RMSE, and MAPE. These metrics offer a thorough evaluation of the model's precision and reliability in forecasting power usage. The formulas for these metrics are as follows:

$$MAE = \frac{1}{n}\sum_{i=1}^{n}|U_i - \widehat{U}_i|, \tag{30}$$

$$MSE = \frac{1}{n}\sum_{i=1}^{n}(U_i - \widehat{U}_i)^2, \tag{31}$$

$$RMSE = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(U_i - \widehat{U}_i)^2}, \tag{32}$$

$$MAPE = \frac{100\%}{n}\sum_{i=1}^{n}\left|\frac{U_i - \widehat{U}_i}{U_i}\right|. \tag{33}$$

here, U_i and $\widehat{U}_i$ are the actual and predicted values, respectively. These performance parameters combined offer a comprehensive assessment of the accuracy and efficacy of the forecasting models in projecting power consumption. MSE and RMSE are effective in identifying significant errors, while MAE and MAPE are easily interpretable measurements of average error and percentage error, respectively.

4.3. Training configuration and validation techniques

This section provides a detailed overview of the architecture and parameters of the models used in our study, including the proposed hybrid model. We also discuss the validation techniques implemented to ensure the robustness and reliability of our results. The configurations were carefully selected to balance model complexity and efficiency, as detailed in Table 2. We compared several models, including LSTM, GRU, BiLSTM, CNN, and our proposed hybrid model, all of which were optimized using various configurations. The depth of the networks varied; the LSTM, GRU, BiLSTM, and CNN architectures were designed with 1 to 4 layers. In contrast, the proposed model featured a deeper architecture with residual connectivity and concatenation layers, allowing it to capture more complex temporal patterns in the data. Each model was configured with varying numbers of neurons per layer, ranging from 50 to 200 for LSTM, BiLSTM, and GRU. For the CNN and hybrid model, the convolutional layers' filters were set to 8, 16, and 32, enabling the model to extract fine-grained and broader features at different scales. The kernel sizes were set to 2 and 3, optimizing the model's ability to detect various patterns in the power consumption data. Training was conducted with batch sizes of 16 or 32, and the models were trained for up to 100 epochs. Early stopping was implemented to halt training if performance on the validation set failed to improve, thus preventing overfitting. A dropout rate of 0.2 was applied to further reduce overfitting. The MSE was selected as the loss function for all models, as it is well-suited for regression tasks such as power consumption forecasting. The Adam optimizer was employed to efficiently minimize the loss function, ensuring the models converged to optimal solutions during training. To assess the generalization ability and robustness of the models, we employed 5-fold cross-validation, which ensured that each model was trained and validated on different subsets of the data. This technique is particularly important in time series forecasting, as it helps mitigate the risks of overfitting by providing a more accurate estimate of model performance across diverse data splits. In addition to cross-validation, the models were evaluated on a separate test set, consisting of 20% of the data, ensuring that the performance metrics were calculated on unseen data. The train-test split of 80% training and 20% testing was maintained across all models for consistency and fair comparison. Furthermore, to enhance the reliability of our results, we tested the proposed hybrid model on two distinct academic building datasets, USB and CIRS, both representing different operational environments. The inclusion of these datasets was essential for testing the model's generalizability. By training and validating datasets from distinct academic buildings, we demonstrated the model's ability to adapt to varying operational patterns and environmental factors, which is crucial for real-world applications in power forecasting. Throughout the evaluation process, we used several performance metrics, such as MAE, MSE, RMSE, and MAPE, to assess the forecasting accuracy of the models. These metrics were calculated for both single-horizon and multi-horizon forecasting tasks, allowing for a comprehensive assessment of the model's performance. The combination of multiple validation techniques, including cross-validation, test set evaluation, and performance metrics, ensures that the proposed hybrid model is both robust and reliable, providing accurate and consistent power consumption forecasts across diverse academic buildings. By employing a rigorous training configuration and a multifaceted validation approach, we address potential concerns regarding the reliability and robustness of the proposed methodology. This ensures that the results presented in this study are

Table 2
Optimal parameter setting of the deep learning models.

Parameters	LSTM	GRU	BiLSTM	CNN	Proposed Model
Depth	1	2	1	4	16 Conv,2 LSTM, 2 MLP layers
Number of neurons per layer	50, 100,200	50, 100	50, 100	–	16, 32 (for LSTM, and MLP layer)
Kernel size	–	–	–	2, 3	2, 3
Number of filter	–	–	–	16, 32	8, 16, 32
Batch size	16, 32				
Epoch	100				
Training set proportion	80%				
Test set proportion	20%				
Loss function	MSE				
Model optimizer	Adam				
Cross validation	5-fold				
Drop-out rate	0.2				
Stopping criteria	Early stopping				

trustworthy and reflective of the model's performance in real-world settings.

4.4. Point forecasting of statistical and data-driven approaches

In this experiment, we evaluated various forecasting approaches, such as SARIMA, Prophet, Persistence, Mean Forecast, and Neural Prophet. We compared the performance of these models with our proposed model, which leverages deep learning techniques.

4.4.1. Statistical approach

In terms of statistical approaches, SARIMA provides a better result than the Prophet algorithm, as shown in Table 8. SARIMA is an extension of the ARIMA model that explicitly supports univariate time series data with a seasonal component. The SARIMA model incorporates seasonality by including additional seasonal terms in the ARIMA model framework. The general form of a SARIMA model is denoted as SARIMA (p, d, q) $(P, D, Q)s$, where p is the order of the autoregressive part, d is the order of differentiation, q is the order of the moving average part, P is the seasonal order of the autoregressive part, D is the seasonal order of differentiation, Q is the seasonal order of the moving average part, s is the seasonal period, indicating the number of time points in a seasonal cycle [45]. In our experiment, we selected the optimal parameters of the SARIMA model using the *auto_arima* function from the pmdarima library, an automated hyperparameter tuning approach. This process automates the model selection process by iterating through various combinations of parameters and selecting the best model based on a specific criterion, such as the Akaike Information Criterion (AIC) and the Bayesian Information Criterion (BIC). For this work, we found the following optimal model parameters: SARIMA (3, 1, 0) (3, 0, 0) 24. Using this hyperparameter configuration, SARIMA achieves the MAE, MSE, RMSE, and MAPE values of 7.11, 91.96, 9.59, and 9.88%, respectively, while the Prophet algorithm achieves 14.15, 309.05, 17.58, and 19.66%, respectively.

4.4.2. Naive baseline methods

In this study, we employed two naive baseline methods: the Persistence algorithm and Mean forecast. The Persistence algorithm is a straightforward yet powerful forecasting technique. It assumes that the subsequent value in the time series will be identical to the last observed value. Despite its simplicity, the Persistence algorithm often serves as a strong baseline for comparison. The Persistence algorithm performed notably well in the experiments, achieving an MAE of 7.35, MSE of 107.53, RMSE of 10.37, and MAPE of 10.05%. Conversely, the Mean forecast method recorded an MAE of 17.42, MSE of 446.89, RMSE of 21.14, and MAPE of 24.55%. The higher error metrics indicate that the Mean forecast method struggles to accurately predict the power consumption patterns in the CIRS building dataset.

4.5. Neural Prophet algorithm

Neural Prophet is an open-source forecasting tool that is intended for time series data with strong seasonal effects and multiple seasonality with daily, weekly, and yearly cycles [47]. Neural Prophet is built to handle missing data and shifts in the trend, making it robust and flexible for various applications. The Neural Prophet is based on an additive model in which non-linear trends are aligned with yearly, weekly, and daily seasonality, as well as holiday effects. The model can be expressed as:

$$y(t) = g(t) + s(t) + h(t) + \epsilon_t \tag{34}$$

where $g(t)$ represents the trend component, modeling non-periodic changes in the time series; $s(t)$ denotes the seasonality component, capturing periodic changes (e.g., daily, weekly, yearly); $h(t)$ accounts for the holiday effects that impact the time series and ϵ_t signifies the error [48]. The Neural Prophet model with default parameters was used in this study for single step prediction using the CIRS building dataset. In this study, Neural Prophet demonstrated robust performance with an MAE of 6.6, MSE of 77.96, RMSE of 8.83 and MAPE of 9.25%. This indicates that the Neural Prophet effectively balances the interpretability of statistical models with the predictive power of machine learning techniques, making it a strong candidate for accurate power consumption forecasting.

However, the results in Table 3 indicate that the proposed model outperforms all other models across all metrics. Specifically, the proposed model achieved the lowest MAE of 5.93, MSE of 71.52, RMSE of 8.46, and MAPE of 7.81%. Among the statistical methods, SARIMA demonstrated superior performance with an MAE of 7.11, MSE of 91.96, RMSE of 9.59, and MAPE of 9.88%. Neural Prophet, as a hybrid model, also performed well with an MAE of 6.6, MSE of 77.96, RMSE of 8.83, and MAPE of 9.25%. In contrast, the Persistence algorithm and Mean forecast models showed comparatively higher error rates, with Mean forecast being the least accurate among all evaluated models.

4.6. Point and multi forecasting of DL techniques

In this section, we present the simulation results of various deep learning models such as LSTM, GRU, CNN, BiLSTM and our proposed model analyzing the CIRS building power consumption dataset in both single and multi-horizon forecasting. Table 4 summarizes deep learning models' performance in predicting the next time step of power consumption in the CIRS building dataset. Each model is evaluated using MAE, MSE, RMSE, and MAPE. LSTM achieved an MAE of 6.57, MSE of 87.47, RMSE of 9.35, and MAPE of 9.31%, indicating strong predictive capability with relatively low error rates. GRU showed slightly higher errors compared to LSTM, with an MAE of 7.04, MSE of 95.77, RMSE of 9.78, and MAPE of 10.23%. Despite this, GRU performed competitively in the single step forecasting task. Designed to capture spatial dependencies, CNN demonstrated an MAE of 8.48, MSE of 141.66, RMSE of 11.90, and MAPE of 11.20%. While generally effective, CNN demonstrated higher error rates compared to LSTM and GRU. BiLSTM, leveraging bidirectional information flow, achieved an MAE of 6.32, MSE of 84.88, RMSE of 9.21, and MAPE of 8.84%, demonstrating robust performance similar to LSTM. However, the proposed model outperformed all other models with an MAE of 5.93, MSE of 71.52, RMSE of 8.46, and MAPE of 7.81%. These results highlight the effectiveness of the proposed model in achieving accurate single-step forecasts, showcasing its potential for enhancing power consumption prediction in academic buildings.

The next experiment presents the results of the multi-step forecasting analysis for a 48-hour horizon using the CIRS building dataset. Four different tables list the simulation results, each representing a different performance metric, i.e., MAE, MSE, RMSE, and MAPE. The MAE values in Table 5 show how well different models do at making multi-step predictions over a 48-hour period using the CIRS building dataset. Each row corresponds to a specific forecasting horizon, ranging from 1 to 48 h. LSTM performs competitively, with MAE values ranging from 6.85 at horizon 1 to 8.67 at horizon 48, demonstrating consistent predictive capability. GRU also performs well, with MAE values ranging from 6.76 to 8.56 across the same horizons. CNN exhibits higher MAE values compared to LSTM and GRU, ranging from 8.89 to 11.33,

Table 3
Performance comparison of statistical and data driven approaches for power consumption forecasting.

Model name	MAE	MSE	RMSE	MAPE (%)
Prophet	14.15	309.05	17.58	19.66
SARIMA	7.11	91.96	9.59	9.88
Neural Prophet	6.6	77.96	8.83	9.25
Mean forecast	17.42	446.89	21.14	24.55
Persistence algorithm	7.35	107.53	10.37	10.05
Proposed Model	**5.93**	**71.52**	**8.46**	**7.81**

Table 4
Performance comparison of deep learning models for single-step forecasting on the CIRS building dataset.

Model name	MAE	MSE	RMSE	MAPE (%)
LSTM	6.57	87.47	9.35	9.31
GRU	7.04	95.77	9.78	10.23
CNN	8.48	141.66	11.90	11.20
BiLSTM	6.32	84.88	9.21	8.84
Proposed Model	**5.93**	**71.52**	**8.46**	**7.81**

indicating slightly lower predictive accuracy. BiLSTM performs similarly to LSTM and GRU in the earlier horizons but shows higher MAE values towards the later horizons, peaking at 10.04. Notably, the proposed model consistently demonstrates robust performance, achieving MAE values ranging from 6.67 to 7.29 across the 48-hour forecasting period, indicating its effectiveness in accurate power consumption prediction. These results underscore the suitability of deep learning models, particularly the proposed model, for effective multi-step forecasting tasks in building power management applications.

The results of the MSE for five deep learning models used for multi-step forecasting using the CIRS building dataset reveal that the proposed model outperforms the others, achieving the lowest MSE values consistently across multiple instances. For example, as shown in Table 6, the proposed model registers an MSE of 77.55 for the sixth horizon, which is significantly lower compared to the LSTM's 117.77, GRU's 114.17, CNN's 175.69, and BiLSTM's 131.21. This trend is observed throughout the dataset, with the proposed model maintaining superior

Table 5
Performance comparison of MAE for multi-horizon testing of CIRS building data.

Horizon	LSTM	GRU	CNN	BiLSTM	**Proposed model**	Horizon	LSTM	GRU	CNN	BiLSTM	**Proposed model**
1	6.85	6.76	8.89	6.72	**6.67**	25	8.06	8.01	10.53	9.22	**6.56**
2	7.00	7.16	9.07	7.80	**6.72**	26	8.20	8.08	10.63	9.34	**6.57**
3	7.50	7.61	9.34	8.22	**6.89**	27	8.23	8.24	10.78	9.65	**6.52**
4	7.59	7.53	9.34	8.03	**6.72**	28	8.25	8.24	10.80	9.47	**6.69**
5	7.50	7.69	9.55	8.12	**6.85**	29	8.22	8.20	10.85	9.63	**6.64**
6	7.88	7.92	9.69	8.28	**6.47**	30	8.41	8.39	11.10	9.62	**6.62**
7	8.02	8.15	10.17	8.57	**6.56**	31	8.33	8.44	11.23	9.68	**6.40**
8	8.01	7.99	10.17	8.85	**6.49**	32	8.39	8.44	11.46	9.68	**6.44**
9	8.05	7.97	9.98	9.19	**6.66**	33	8.20	8.34	11.17	9.61	6.45
10	7.97	7.97	10.07	8.79	**6.48**	34	8.38	8.42	11.30	9.61	**6.57**
11	8.18	8.06	10.04	8.75	**6.55**	35	8.58	8.33	11.36	9.57	**6.44**
12	8.01	8.11	9.98	8.93	**6.62**	36	8.58	8.28	11.25	9.67	**6.44**
13	8.04	8.16	10.07	8.98	**6.54**	37	8.38	8.22	11.48	9.61	**6.44**
14	7.87	8.12	10.15	8.86	**6.42**	38	8.22	8.33	11.30	9.56	**6.47**
15	7.93	8.16	10.15	8.89	**6.64**	39	8.53	8.45	11.58	9.76	**6.63**
16	7.87	8.01	10.14	8.80	**6.51**	40	8.55	8.32	11.49	9.66	**6.71**
17	7.90	7.94	10.04	8.90	**6.41**	41	8.40	8.14	11.34	9.63	**6.79**
18	7.81	7.89	10.12	8.79	**6.36**	42	8.57	8.16	11.26	9.58	**6.74**
19	7.76	7.90	9.96	8.85	**6.33**	43	8.33	8.12	11.01	9.70	**6.60**
20	7.66	7.75	9.76	8.60	**6.31**	44	8.39	8.20	11.02	9.55	**6.69**
21	7.85	7.87	9.88	8.76	**6.46**	45	8.56	8.30	10.93	9.67	**6.73**
22	7.74	7.78	9.92	8.69	**6.45**	46	8.41	8.31	11.01	9.56	**6.92**
23	8.01	8.00	10.29	8.91	**6.58**	47	8.69	8.37	11.12	9.83	**6.92**
24	7.91	7.93	10.50	8.96	**6.39**	48	8.67	8.56	11.33	10.04	**7.29**

Table 6
Performance comparison of MSE for multi-horizon testing of CIRS building data.

SN	LSTM	GRU	CNN	BiLSTM	**Proposed model**	SN	LSTM	GRU	CNN	BiLSTM	**Proposed model**
1	84.15	92.44	149.44	87.41	**91.76**	25	122.98	119.30	202.09	153.40	**80.33**
2	96.59	100.70	157.49	111.51	**85.20**	26	125.97	120.61	205.95	156.65	**78.62**
3	105.64	108.64	166.21	121.96	**88.99**	27	124.39	123.88	211.47	165.78	**80.70**
4	104.92	107.00	163.51	118.86	**80.08**	28	126.40	124.20	210.80	162.88	**80.40**
5	106.62	110.05	171.62	125.65	**82.55**	29	128.67	125.28	212.44	166.14	**80.76**
6	117.77	114.17	175.69	131.21	**77.55**	30	128.55	129.00	217.62	166.66	**80.83**
7	121.83	119.62	191.50	136.28	**80.97**	31	127.43	129.72	223.96	168.48	**75.80**
8	121.19	118.63	189.37	143.46	**79.34**	32	127.93	131.67	232.12	170.12	**79.24**
9	121.81	120.82	185.69	149.25	**79.25**	33	123.10	125.96	224.00	165.19	**78.44**
10	120.23	120.25	189.17	143.30	**77.15**	34	127.15	132.30	228.09	168.81	**82.91**
11	125.82	123.80	189.37	144.43	**81.44**	35	130.64	130.38	229.17	169.05	**80.12**
12	119.20	121.89	188.11	146.12	**78.19**	36	130.27	129.06	226.34	169.92	**77.49**
13	119.55	122.46	191.39	147.03	**78.95**	37	126.95	125.80	231.18	167.56	**80.45**
14	116.09	124.27	193.39	147.03	**77.80**	38	124.13	129.46	228.76	165.47	**79.64**
15	120.34	127.69	195.87	149.14	**84.52**	39	135.02	137.10	238.16	175.01	**86.13**
16	115.91	122.56	193.81	144.90	**82.17**	40	134.24	130.98	236.51	172.84	**84.69**
17	114.34	120.36	187.89	143.19	**78.11**	41	129.99	125.61	225.81	167.92	**86.25**
18	110.69	114.32	188.54	139.70	**75.24**	42	129.16	123.68	221.88	166.66	**84.71**
19	113.25	115.97	185.00	143.61	**76.91**	43	127.02	121.42	212.87	165.12	**82.53**
20	107.63	111.56	179.23	135.78	**74.02**	44	131.05	127.45	215.38	166.00	**85.51**
21	112.15	115.39	183.43	144.38	**76.36**	45	133.51	131.95	214.18	170.25	**86.33**
22	110.79	114.06	185.78	140.66	**75.93**	46	130.68	133.13	217.38	168.00	**88.65**
23	116.45	121.94	196.27	147.30	**80.48**	47	133.32	132.98	218.77	170.11	**91.60**
24	115.29	118.56	198.63	144.98	**75.17**	48	138.17	139.45	224.47	175.89	**101.01**

performance. The combination of convolutional layers, residual connections, concatenation, factorization, and LSTM layers in the proposed model enables it to capture both spatial and temporal dependencies effectively, leading to better accuracy. The LSTM model also demonstrates competitive performance due to its capability to handle long-term dependencies in sequential data, while the GRU model, despite its simpler architecture, performs slightly worse than LSTM. For example, the LSTM model has strong performance, with MSE values ranging from 84.15 at horizon 1 to 138.17 at horizon 48, which indicates the model's stability in making predictions. Similarly, GRU obtained MSE values ranging from 92.44 to 139.45. The CNN model exhibits the highest MSE values, highlighting its limitation in capturing temporal dependencies as effectively as LSTM or GRU. The BiLSTM model, although beneficial in

understanding context through bidirectional processing, has slightly higher MSE values due to potential overfitting or training complexities. Therefore, CNN and BiLSTM show higher MSE values compared to LSTM and GRU, suggesting slightly reduced predictive accuracy over longer forecasting horizons. Overall, the cleverly designed strategies of the proposed model, which integrates various advanced features, allow for the most accurate power consumption forecasting by achieving decreased MSE values for all time periods, ranging from 74.02 to 101.01 in this study.

Table 7 lists the RMSE values obtained from the five DL models that were evaluated for multi-step forecasting on the CIRS building dataset. RMSE provides a measure of the average magnitude of error in predictions, considering both the magnitude and direction of error. The LSTM and GRU models demonstrate comparable performance, with RMSE values ranging from 9.17 to 11.75 and 9.61 to 11.80, respectively, across various forecasting timeframes. CNN and BiLSTM have significantly higher RMSE values than LSTM and GRU, indicating a notable decrease in accuracy in capturing temporal variability. For instance, the lowest RMSE value for BiLSTM and CNN is 9.29 and 12.22, and the highest value is 13.47 and 15.43, respectively. However, like the MSE, the proposed model attains the lowest RMSE values across different forecast horizons. For instance, at horizon 1, the proposed model achieves an RMSE of 9.23, while LSTM, GRU, CNN, and BiLSTM record RMSEs of 9.27, 9.61, 12.22, and 9.29, respectively. This trend continues as the forecast horizon increases, with the proposed model maintaining superior performance, such as an RMSE of 8.96 at horizon 25 compared to LSTM's 11.08, GRU's 10.92, CNN's 14.21, and BiLSTM's 12.28.

The MAPE results for the deep learning models in forecasting the power consumption of the CIRS building dataset reveal varying levels of accuracy across different horizons, as shown in Table 8. The proposed model demonstrates outstanding performance for short-term forecasting with the least MAPE among all other DL models. For example, at the 14th horizon, the MAPE value for the proposed model, GRU, BiLSTM, LSTM, and CNN is 8.73%, 11.39%, 12.30%, 12.70%, and 13.17%, respectively. This trend persists across multiple horizons. For instance, at 25th horizon, the proposed model maintains its superiority with a MAPE of 9.07%, which is notably lower than the MAPE values of LSTM (11.76%), GRU (11.73%), CNN (13.83%), and BiLSTM (13.01%). The proposed model's architecture, which integrates convolutional layers, residual connections, concatenation, factorization, and LSTM layers, allows it to effectively capture both spatial and temporal dependencies, leading to more accurate predictions. Overall, the robust architecture of the proposed model allows it to achieve the lowest MAPE values across various horizons, highlighting its efficacy in accurately forecasting power consumption. The LSTM model also performs well, due to its capacity to handle long-term dependencies. The GRU and BiLSTM models, although effective, do not match the proposed model's accuracy, while the CNN model's limitations in time-series forecasting are evident in its higher MAPE values.

4.7. Performance evaluation with USB building dataset

To assess the robustness and generalizability of our proposed model, we conducted additional experiments using a new dataset known as the USB Building dataset. This secondary evaluation aims to validate the performance of the models in a different context and ensure their effectiveness across diverse datasets. We conducted experiments for both single-horizon and multi-horizon forecasting using the LSTM, GRU, CNN, BiLSTM, and the proposed model. This section will discuss the performance results when using the USB building dataset. The performance comparison of deep learning models for single step forecasting on the USB Building dataset demonstrates the superior effectiveness of the proposed model.

As shown in the Table 9, the proposed model achieves the lowest MAE of 5.93, MSE of 78.52, RMSE of 8.86, and MAPE of 2.60%. In contrast, the LSTM model records higher error values with an MAE of

Table 7
Performance comparison of RMSE for multi-horizon testing of CIRS building data.

SN	LSTM	GRU	CNN	BiLSTM	**Proposed model**	SN	LSTM	GRU	CNN	BiLSTM	**Proposed model**
1	9.27	9.61	12.22	9.29	**9.23**	25	11.08	10.92	14.21	12.28	**8.96**
2	9.82	10.03	12.54	10.19	**9.35**	26	11.22	10.98	14.35	12.67	**8.86**
3	10.27	10.42	12.89	10.98	**9.43**	27	11.15	11.13	14.54	12.69	**8.98**
4	10.24	10.34	12.78	10.99	**8.94**	28	11.24	11.14	14.51	12.63	**8.96**
5	10.32	10.49	13.10	11.23	**9.08**	29	11.34	11.19	14.57	12.74	**8.98**
6	10.85	10.68	13.25	11.58	**8.80**	30	11.33	11.35	14.75	12.98	**8.99**
7	11.03	10.93	13.83	11.81	**8.99**	31	11.28	11.38	14.96	13.14	**8.70**
8	11.00	10.89	13.76	11.89	**8.90**	32	11.31	11.47	15.23	13.05	**8.90**
9	11.03	10.99	13.62	12.23	**8.90**	33	11.09	11.22	14.96	13.05	**8.85**
10	10.96	10.96	13.75	12.23	**8.78**	34	11.27	11.50	15.10	13.08	**9.10**
11	11.21	11.12	13.76	11.98	**9.02**	35	11.43	11.41	15.13	12.95	**8.95**
12	10.91	11.04	13.71	12.05	**8.84**	36	11.41	11.36	15.04	12.77	**8.80**
13	10.93	11.06	13.83	12.32	**8.88**	37	11.26	11.21	15.20	12.98	**8.96**
14	10.77	11.14	13.9	12.24	**8.82**	38	11.14	11.37	15.12	12.93	**8.92**
15	10.97	11.30	13.99	12.32	**9.19**	39	11.61	11.70	15.43	13.47	**9.28**
16	10.76	11.07	13.92	12.15	**9.06**	40	11.58	11.44	15.37	12.86	**9.20**
17	10.69	10.97	13.7	12.02	**8.83**	41	11.40	11.20	15.02	12.58	**9.28**
18	10.52	10.69	13.73	12.29	**8.67**	42	11.36	11.12	14.89	12.65	**9.20**
19	10.64	10.76	13.6	12.23	**8.60**	43	11.27	11.01	14.59	12.53	**9.08**
20	10.37	10.56	13.38	12.30	**8.73**	44	11.44	11.28	14.67	12.65	**9.24**
21	10.59	10.74	13.54	12.37	**8.71**	45	11.55	11.48	14.63	12.66	**9.29**
22	10.52	10.68	13.63	11.78	**8.97**	46	11.43	11.53	14.74	12.79	**9.41**
23	10.79	11.04	14.0	12.04	**8.67**	47	11.54	11.53	14.79	12.77	**9.57**
24	10.73	10.88	14.09	11.95	**8.84**	48	11.75	11.80	14.98	13.05	**10.05**

Table 8
Performance comparison of MAPE for multi-horizon testing of CIRS building data.

SN	LSTM %	GRU %	CNN %	BiLSTM %	**Proposed model (%)**	SN	LSTM %	GRU %	CNN %	BiLSTM %	**Proposed model (%)**
1	9.22	9.40	11.42	9.38	**9.16**	25	11.76	11.73	13.83	13.01	**9.07**
2	9.67	10.01	12.16	11.11	**9.20**	26	11.96	11.59	13.97	13.08	**9.26**
3	11.16	10.88	13.39	11.67	**9.47**	27	11.99	11.79	14.54	13.70	**9.01**
4	10.45	10.75	13.40	11.20	**9.34**	28	11.90	11.93	15.29	13.08	**9.54**
5	10.87	11.20	13.21	11.18	**9.69**	29	12.17	11.89	15.19	13.70	**9.39**
6	11.25	11.11	13.52	11.20	**8.87**	30	12.34	12.24	15.24	13.38	**9.29**
7	11.40	11.55	13.33	11.83	**9.09**	31	12.40	12.21	15.12	13.45	**8.91**
8	11.57	12.11	13.20	12.36	**8.90**	32	12.65	12.24	14.93	13.39	**8.91**
9	12.76	11.78	13.40	13.07	**9.36**	33	12.98	11.99	14.96	13.53	**8.99**
10	12.18	11.60	13.66	12.22	**9.00**	34	13.14	12.11	15.47	13.26	**8.80**
11	12.78	11.46	13.11	11.88	**9.03**	35	12.89	12.22	14.93	13.30	**8.74**
12	13.02	11.62	13.65	12.47	**9.29**	36	12.44	12.15	15.47	13.56	**8.89**
13	12.74	11.66	13.49	12.56	**9.00**	37	12.41	12.03	15.65	13.38	**8.72**
14	12.70	11.39	13.17	12.30	**8.73**	38	12.43	11.97	15.35	13.39	**8.80**
15	12.20	11.69	13.22	12.23	**8.99**	39	12.14	12.16	15.41	13.60	**9.02**
16	11.95	11.39	13.28	12.19	**8.90**	40	11.99	12.16	15.26	13.25	**9.17**
17	11.87	11.25	13.14	12.41	**8.86**	41	12.36	12.01	15.15	13.28	**9.23**
18	11.88	11.35	13.33	11.99	**8.73**	42	12.23	12.07	14.89	13.14	**9.28**
19	11.58	11.29	13.50	8.63	**8.63**	43	12.05	12.00	14.35	13.75	**9.04**
20	11.45	11.05	13.11	11.84	**8.69**	44	12.50	12.07	14.97	13.27	**9.19**
21	11.92	11.34	13.41	11.87	**9.05**	45	12.54	11.94	14.66	13.50	**9.23**
22	11.95	11.44	13.15	12.05	**9.07**	46	12.67	11.94	14.55	13.36	**9.68**
23	11.64	11.62	13.37	12.20	**9.26**	47	13.28	12.00	14.31	13.86	**9.34**
24	11.74	11.59	13.65	12.77	**8.88**	48	13.46	12.33	14.89	14.21	**9.90**

Table 9
Performance comparison of deep learning models for single-step forecasting on the USB building dataset.

Model	MAE	MSE	RMSE	MAPE (%)
LSTM	9.75	213.95	14.62	4.29
GRU	10.62	231.91	15.22	4.67
CNN	16.10	516.20	22.72	7.08
BiLSTM	9.65	204.11	14.29	3.11
Proposed Model	**5.93**	**78.52**	**8.86**	**2.60**

9.75, MSE of 213.95, RMSE of 14.62, and MAPE of 4.29%. The GRU model performs slightly worse than the LSTM, with an MAE of 10.62, MSE of 231.91, RMSE of 15.22, and MAPE of 4.67%. The CNN model exhibits the highest error metrics among all models, indicating its lesser suitability for this task, with an MAE of 16.10, MSE of 516.20, RMSE of 22.72, and MAPE of 7.08%. The BiLSTM model shows competitive performance but still falls short of the proposed model, with an MAE of 9.65, MSE of 204.11, RMSE of 14.29, and MAPE of 3.11%. The results demonstrate the efficient performance of the proposed model in unit-step forecasting on the USB Building dataset. The proposed model outperforms traditional LSTM, GRU, CNN, and BiLSTM models by a significant margin.

This section evaluates the performance of the DL models in terms of the MAE, MSE, RMSE in multi-step forecasting of power consumption. Fig. 3(a) reveals that the proposed hybrid model consistently achieves the lowest MAE values, ranging from 13.63 to 16.15, indicating its superior accuracy and robustness. The LSTM and GRU models also perform well, with MAE values ranging from 13.70 to 20.15 and 14.59 to 18.43, respectively, leveraging their ability to capture long-term dependencies in time series data. The BiLSTM model performs slightly better than the standard LSTM and GRU, with MAE values between 14.17 and 18.36, benefiting from its bidirectional nature to consider past and future contexts. In contrast, the CNN model, with MAE values ranging from 21.42 to 27.68, struggles with temporal dependencies, highlighting its limitation in power consumption forecasting despite its strength in spatial feature extraction.

The same observation was made in the case of MSE and RMSE for validating the USB building dataset at 48 step prediction, as shown in Fig. 3(b) and Fig. 3(c). With the lowest MSE values, the proposed model outperforms all other models, demonstrating its unparalleled accuracy and robustness in power consumption forecasting. For example, at the 4th horizon, the proposed model's MSE value is 397.76, while the MSE values for LSTM, BiLSTM, GRU, and CNN are 546.84, 547.19, 557.83, and 1006.9, respectively. However, the LSTM and GRU models also demonstrate good performance. For instance, LSTM achieves MSE values ranging between 254.69 and 843.08, and the GRU shows MSE of

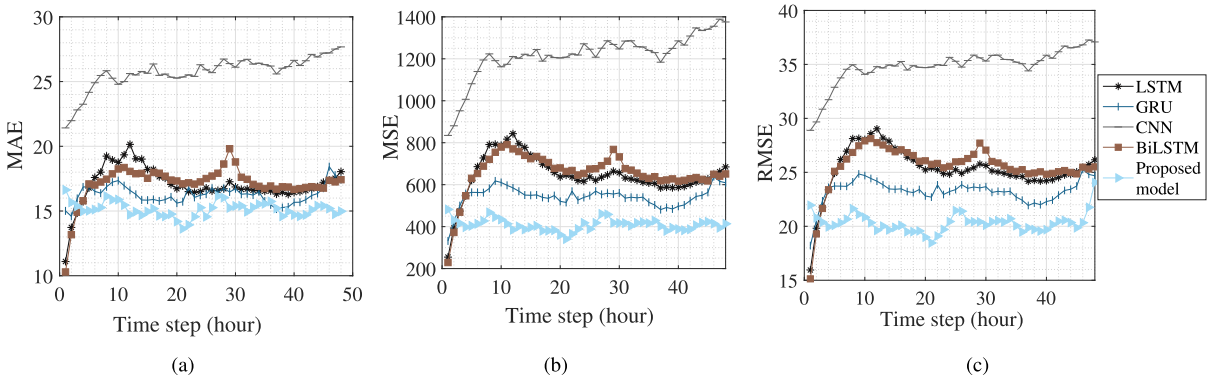


Fig. 3. Performance comparison of various DL models for multi-horizon testing on USB building dataset: (a) MAE, (b) MSE, and (c) RMSE results.

331.80 to 638.41. The BiLSTM model exhibits a marginally superior performance compared to the LSTM and GRU models, as evidenced by the MSE values ranging from 228.98 to 791.24. The CNN model, exhibiting MSE values between 834.77 and 1389.00, faces challenges in capturing temporal dependencies, underscoring its limitation in power consumption forecasting despite its proficiency in spatial feature extraction.

Fig. 4 compares the actual and predicted power consumption of the proposed model with other DL models. This figure shows the actual and predicted values for multistep (48 horizons) forecasting for one of the sequence samples (index 4000). The comparison is based on the USB building dataset for multi-horizon power consumption forecasting. The forecasting outcomes for a range of deep learning models, including LSTM, GRU, CNN, BiLSTM, and the proposed model, are presented in the subfigures, Fig. 4(a), Fig. 4(b), Fig. 4(c), Fig. 4(d), and Fig. 5(e), respectively. The proposed model, displayed in subfigure (e), consistently delivers predictions closely aligned with the actual values, highlighting its superior performance compared to the other models.

4.8. Computational complexity analysis of DL models

Fig. 5 shows the computational complexity analysis for DL models, including LSTM, GRU, CNN, and BiLSTM, for single-step forecasting in CIRS and USB buildings. The GRU model exhibited the highest number of learnable parameters and memory requirements, 101101, and 394.93 kB, respectively, reflecting its complexity. It also has the longest inference time (16.97 seconds for CIRS and 18.53 seconds for USB). The LSTM and BiLSTM models presented moderate complexity, with 50401 and 30401 parameters, and 196.88 kB and 118.75 kB memory size, respectively. The inference times of LSTM and BiLSTM were longer than CNN (7.16 and 8.414 seconds for LSTM; 10.15 and 12.01 seconds for BiLSTM). The CNN model demonstrated the most computational efficiency with the least number of parameters (1713), and the least memory usage (6.69 kB), resulting in the fastest inference time (0.624 seconds for CIRS and 0.589 seconds for USB). However, the CNN model's forecasting accuracy is lowest compared to the model discussed in the previous sections. In contrast, the learnable parameters, memory, and the inference time for the CIRS and USB building for the proposed model are 20, 257, 79.13 kB, 1.9 and 1.7 seconds, respectively. The proposed model has a slightly higher computational expense compared to the CNN model, but it achieves higher forecasting accuracy than the mentioned DL models, including CNN.

Fig. 6 shows the computational complexity analysis for multi-step (48 horizon) forecasting in the CIRS and USB buildings, revealing distinct differences in model performance and resource demands. Similar to the single step, the GRU model has the highest learnable parameters (552K) and memory requirement (2.11 MB), leading to significant computational demand and the longest inference times (16.64 seconds for CIRS and 17.92 seconds for USB). In contrast, the CNN model stands out for its efficiency with the least parameters (5K), minimal memory usage (0.021 MB), resulting in the fastest inference times (0.612 seconds for CIRS and 0.63 seconds for USB). However, this model is inappropriate for the building power forecasting system due to the low prediction accuracy. The BiLSTM model has lower learnable parameters, and memory than the GRU and LSTM models. The learnable parameters, and memory for the BiLSTM and the LSTM model are 481K, and 1.84 MB; and 501K, and 1.91 MB, respectively. Nevertheless, the LSTM model's inference time is less than that of the BiLSTM and GRU models in the CIRS building (7.3 seconds) and USB building (6.5 seconds) datasets. However, the proposed model shows remarkable results in terms of both forecasting accuracy and computational expenses. The proposed model, with 21K learnable parameters and 0.0822 MB of memory, strikes a balance between computational efficiency and performance. It shows faster inference times (1.334 seconds for CIRS and 1.658 seconds for USB), making it an attractive option for scenarios that require efficient yet effective multi-step forecasting.

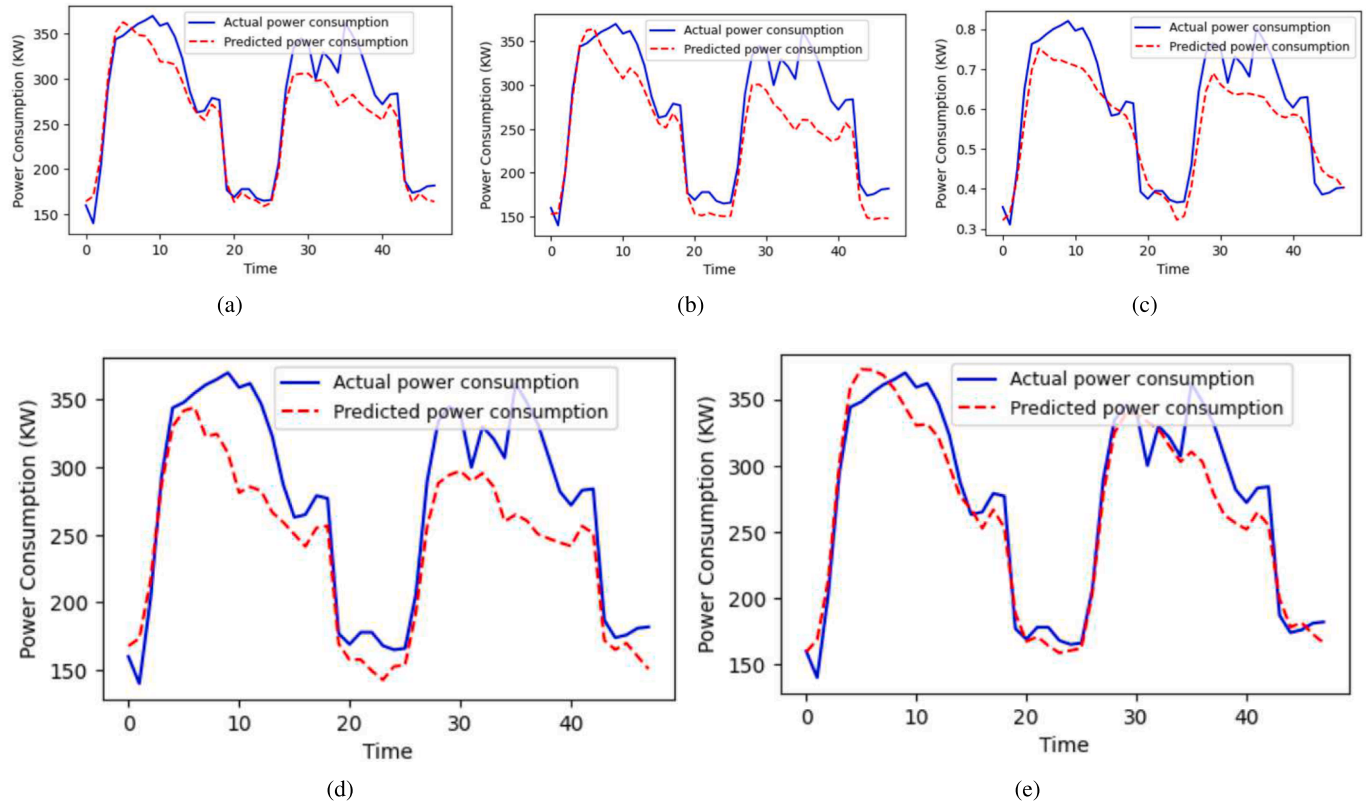


Fig. 4. Performance comparison of actual vs. predicted power consumption using USB building dataset in multi-horizon forecasting for (a) LSTM, (b) GRU, (c) CNN, (d) BiLSTM, and (e) Proposed model.

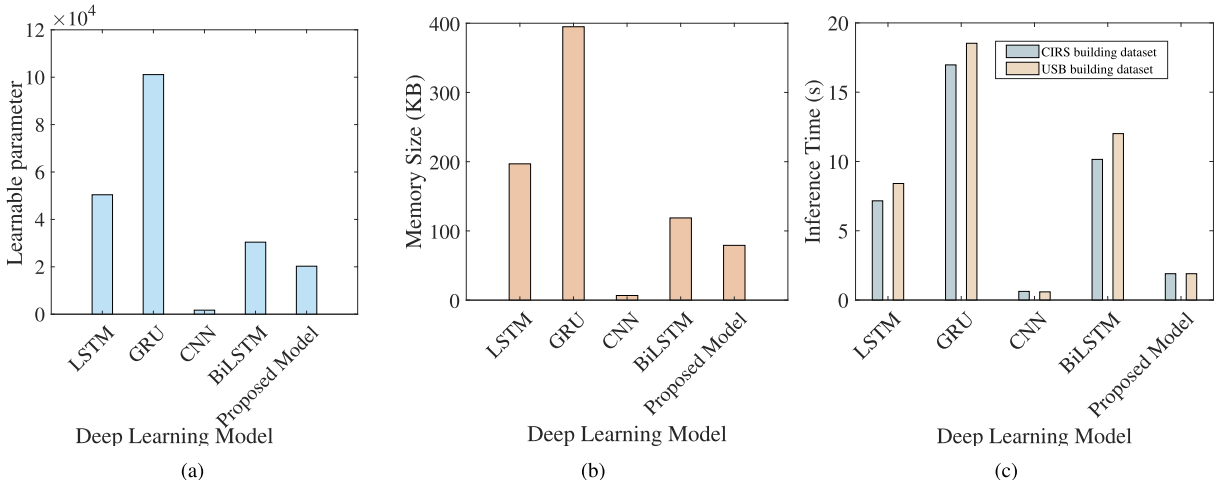


Fig. 5. Computational complexity analysis of DL models for single-step forecasting: (a) Learnable parameters, (b) Memory size, and (c) Inference time.

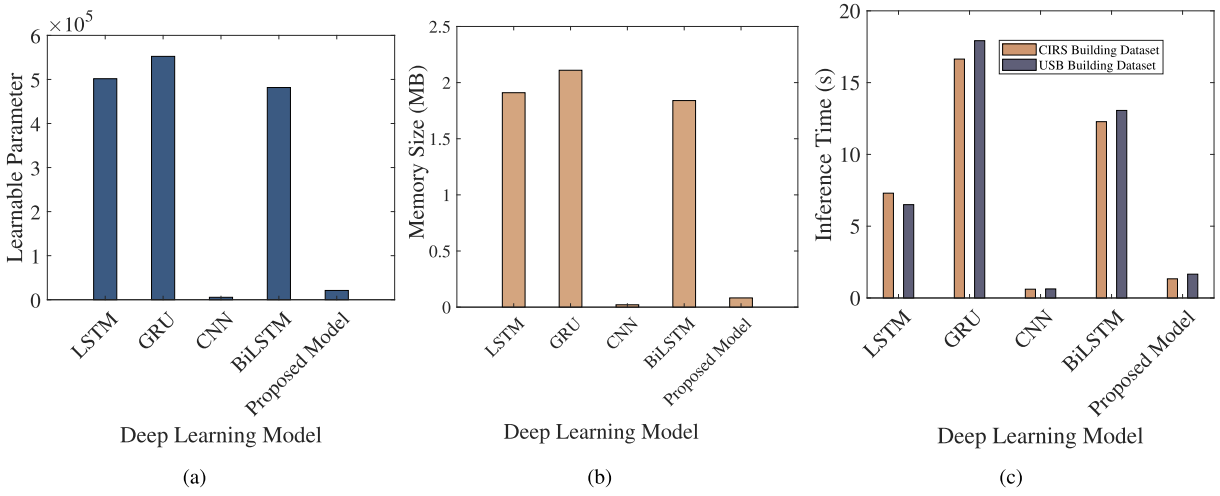


Fig. 6. Computational complexity analysis of DL models for multi-step forecasting: (a) Learnable parameters, (b) Memory size, and (c) Inference time.

4.9. Ablation study of the proposed model

To better understand the contributions of various critical components in our hybrid deep learning model, we performed an ablation study systematically altering key architectural elements. This study was carried out using CIRS building datasets for multi-horizon forecasting tasks. The evaluation metrics employed included MAE, MSE, RMSE, and MAPE. We also considered the number of learnable parameters and inference time as measures of computational complexity. Table 10 shows the detailed structure of the network, along with various configurations, which are listed below.

- Model-1 is designed with LSTM and MLP layers arranged in a sequential manner. It acts as a baseline model to analyze the individual contributions of both components.
- Model-2 consists of two blocks of LSTM layers, where the first LSTM layer has 32 neurons and the second has 16 neurons. This structure eliminates the Conv-block and the MLP-block, allowing us to evaluate the performance of LSTM-only networks in forecasting.
- Model-3 incorporates the Conv-block, MLP-block, and all the residual and concatenation layers following the proposed model architecture. This model eliminates the LSTM-block from the baseline structure.

Table 10
Various network structures of the proposed model.

Model	Conv-block	LSTM-block	MLP-block	Residual connection	Concatenation
Proposed model	2 Conv-block	2 LSTM	2 MLP-block	✓	✓
Model-1	×	✓	✓	×	×
Model-2	×	✓	×	×	×
Model-3	✓	×	✓	✓	✓
Model-4	✓	✓	×	✓	✓
Model-5	3 Conv-block	✓	✓	✓	✓
Model-6	✓	3 LSTM-block	✓	✓	✓
Model-7	✓	✓	3 MLP-block	✓	✓
Model-8	Sequential Conv layer	✓	✓	×	✓
Model-9	Sequential Conv layer	✓	✓	×	×

- Model-4 is built with the Conv-block and the LSTM-block, omitting the MLP-block from the baseline model, while other network configurations remain the same.

- Model-5, Model-6, and Model-7 each feature three blocks of Conv, LSTM, and MLP, respectively, while the proposed baseline model comprises two blocks of each key unit. This configuration showcases the performance of more complex architecture and evaluates the effectiveness of adding incremental blocks to capture sequential dependencies.
- Model-8 and Model-9 consist of two Conv-block, LSTM-block, and MLP-block. Instead of residual connectivity, the Conv-block is constructed by stacking the convolution layer sequentially. Notably, Model-8 employed the concatenation layer in the sequential convolution layer, whereas it is eliminated in Model-9.

The ablation study presented here aims to explore the contributions of individual components within our proposed deep learning architecture for power consumption forecasting. These analyses provide valuable insights into the effectiveness of different design choices in optimizing the model's accuracy and computational efficiency. As shown in Table 11, Model-1 performance deteriorates than the proposed model when we remove the Conv-block. MAE, MSE, RMSE, and MAPE all increase, indicating that the Conv-block plays a vital role in improving forecasting accuracy by capturing important features from the input data. Despite having fewer parameters (15,072) and a higher inference time (2.98 s) than the proposed model, Model-1 suffers from reduced generalization capability. This result shows that the inclusion of Conv-block is critical for extracting meaningful hierarchical patterns in the data. Eliminating both the Conv-block and the MLP-block in Model-2 results in a significant drop in performance. The increased error metrics indicate that relying solely on LSTM layers, without the support of additional blocks like Conv and MLP, leads to less accurate predictions. Moreover, the higher number of learnable parameters (160,176) than the proposed model adds complexity to the model and further hampers its generalization ability. Additionally, the inference time (16.41 s) is longer compared to the proposed model, suggesting that using LSTM alone is inadequate for efficient forecasting without the enhanced feature extraction and transformation capabilities provided by convolutional and MLP layers. In Model-3, removing the LSTM block resulted increase in MAE, MSE, RMSE, and MAPE compared to the proposed model. However, with only 12,536 learnable parameters, Model-3 is more computationally efficient and achieves a faster inference time (0.96 s) than the proposed model. Despite this efficiency, the absence of LSTM blocks limits the model's ability to capture long-term dependencies in the time-series data, reducing the accuracy of its predictions. This suggests that LSTM layers play a crucial role in modeling temporal relationships within the data. Nevertheless, the model can still perform reasonably well by utilizing Conv and MLP layers, even without LSTM blocks. When the MLP block is removed, as discussed in Model-4, the model's performance decreases substantially compared to the proposed model. Metrics such as MAE, MSE, RMSE, and MAPE increase, indicating that the MLP block positively contributes to prediction accuracy, particularly in capturing nonlinear patterns and refining feature representation. Having fewer learnable parameters (14,280), this model remains computationally efficient, with a very low inference time of 0.48 s. This emphasizes the role of MLP layers in optimizing the model's ability to map learned features into meaningful outputs, though the model still struggles without the MLP block.

Table 11
Performance comparison across different architectures of the proposed model.

Model	MAE	MSE	RMSE	MAPE (%)	Learnable parameter	Inference time (Sec.)
Proposed model	6.65	82.61	9.08	9.18	21,056	1.33
Model-1	8.91	146.6	12.11	12.29	15,072	2.98
Model-2	9.12	157.54	12.55	12.37	160,176	16.41
Model-3	8.86	144.16	12	12.1	12,536	0.96
Model-4	9.19	155.11	12.45	12.57	14,280	0.48
Model-5	6.83	87.16	9.33	9.4	27,568	1.82
Model-6	6.35	77.86	8.82	8.73	29,376	2.18
Model-7	7.02	89.65	9.46	9.82	22,112	1.17
Model-8	10.89	219.61	14.81	14.94	20,464	25.85
Model-9	8.83	144.99	12.04	12.06	15,360	1.35

The ablation study on Model-5, Model-6, and Model-7, which are derived by incrementally adding one block to the proposed model's architecture, reveals key insights into the contribution of each block type to the overall performance. Model-5 demonstrates a increase in forecasting errors, with MAE rising to 6.83 and RMSE to 9.33, suggesting that while the Conv-block can capture more local features, the additional Conv-block may lead to overfitting and increased complexity, as reflected by the higher number of learnable parameters (27,568) and a longer inference time (1.82 s). In contrast, Model-6 outperforms all other variants, achieving the lowest MAE of 6.35, RMSE of 8.82, and MAPE of 8.73. This indicates that the added LSTM layers enhance the model's ability to capture sequential dependencies, leading to more accurate time-series forecasting. Although it comes with a higher number of learnable parameters (29,376) and increased inference time (2.18 s), the improvements in accuracy validate the value of deeper LSTM architectures for complex temporal patterns. Model-7 shows an increase in forecasting errors, with MAE of 7.02 and RMSE of 9.46, suggesting that while the additional MLP layers contribute to learning non-linearities, they do not enhance the model's ability to capture time-dependent features compared to the LSTM-based model. The increase in learnable parameters (22,112) and relatively lower inference time (1.17 s) also points to a trade-off between model complexity and performance, underscoring the importance of balanced block integration. Overall, these results highlight the nuanced impact of adding layers in different network components.

The ablation study results for Model-8 and Model-9, which investigate the impact of removing residual connections and the concatenation layer, provide significant insights into the importance of these architectural components for model performance. Model-8, which omits the residual connections from the Conv-block, demonstrates a substantial decline in forecasting accuracy, as evidenced by the sharp increase in MAE (10.89), MSE (219.61), RMSE (14.81), and MAPE (14.94). The model's performance degradation suggests that the residual connections play a crucial role in stabilizing the learning process, helping mitigate vanishing gradients and promoting more efficient gradient flow, particularly in deep networks. Although the number of learnable parameters is relatively low (20,464) and inference time is considerably longer (25.85 s), the absence of residual connections leads to a significant increase in the model's inability to capture the temporal dependencies and complex patterns within the data. In contrast, Model-9, which removes both the residual and concatenation connections, shows improved performance compared to Model-8 but still lags the proposed model. With MAE of 8.83, MSE of 144.99, and RMSE of 12.04, this model performs better than Model-8, indicating that while the concatenation layer is beneficial, its absence does not lead to as drastic a decline in performance as the removal of residual connections. The MAPE of 12.06 is still relatively high, indicating that although concatenation aids in combining feature information, the residual connection significantly enhances learning and prediction accuracy. Furthermore, Model-9 has fewer learnable parameters (15,360) and a similar inference time (1.35 s) compared to the proposed model. However, its performance shows that the interaction between residual and concatenation connections provides essential information flow within the network, helping the model learn more effectively. Overall, the results emphasize the importance of residual connections and concatenation layers in enhancing the model's ability to learn long-term dependencies and produce more accurate forecasts. Removing these components leads to poorer performance, especially in terms of generalization and convergence speed. However, the proposed model achieves outstanding results compared to other network configurations, highlighting its effectiveness. This is attributed to the combination of two Conv-block, two LSTM-blocks, and two MLP-block, all enhanced

with residual connections and concatenation layers.

5. Conclusion

The proposed hybrid model, integrating CNN, LSTM, and MLP components with residual connectivity and concatenation, demonstrates superior performance in multi-horizon power load forecasting for academic buildings. By leveraging the strengths of each component, the model effectively captures both spatial and temporal dependencies, addressing the complex and variable power consumption patterns inherent in academic settings. Residual connectivity ensures efficient training of deeper networks, while concatenation and the three-block approach enhance the model's capability to manage diverse load dynamics and improve computational efficiency. In simulations, we investigate the robustness of the proposed model using two distinct sets of academic building datasets. The performance of the proposed model is compared with the statistical methods, Naive baseline methods, Neural Prophet, and other DL models in terms of MSE, MAE, RMSE, and MAPE. Moreover, the computational complexity of the DL models is also assessed by learnable parameters, memory size, and inference time. Our extensive evaluation shows that the proposed model consistently achieves lower MAE, MSE, RMSE, and MAPE values and outperforms all the other models considered here. This underscores its robustness and accuracy in handling single and multi-horizon forecasting tasks. Furthermore, the model's ability to integrate exogenous variables and adapt to the unique characteristics of various academic buildings enhances its practical applicability and generalizability. Overall, the proposed model offers a significant advancement in the field of power load forecasting, providing a reliable and efficient solution for long-term power management and operational planning in academic institutions. The proposed model can be enhanced in future work by utilizing multi-task learning, enabling the simultaneous forecasting of power consumption, anomaly detection, peak demand prediction, and other related tasks across different building types. Additionally, the model's scalability could be improved by incorporating multivariate datasets, which would capture a wider range of influencing factors, such as occupancy and environmental conditions, across a larger number of buildings. This enhancement would improve the model's ability to generalize to diverse scenarios and boost its forecasting performance. Furthermore, the exploration of real-time forecasting and the integration of external factors, such as weather conditions and energy prices, could significantly increase the model's practical applicability and accuracy.

CRediT authorship contribution statement

Rubina Akter: Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Majid Gholami Shirkoohi:** Writing – review & editing, Validation, Software, Methodology, Formal analysis, Conceptualization. **Jing Wang:** Writing – review & editing, Writing – original draft. **Walter Mérida:** Writing – review & editing, Supervision, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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